

2026 ESC Guidelines for the management of heart failure

Official Slide Set

Slide set history

The material was adapted from the '*2026 ESC Guidelines for the management of heart failure*' (*European Heart Journal*; 2026 - doi: 10.1093/eurheartj/ehag100) as published on 28/08/26.

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ESC Classes of recommendations

Class of recommendation	Definition	Wording to use
I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	<i>“is recommended” or “is indicated”</i>
II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	<i>“should be considered”</i>
IIb	Usefulness/efficacy is less well established by evidence/opinion.	<i>“may be considered”</i>
III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	<i>“is not recommended”</i>

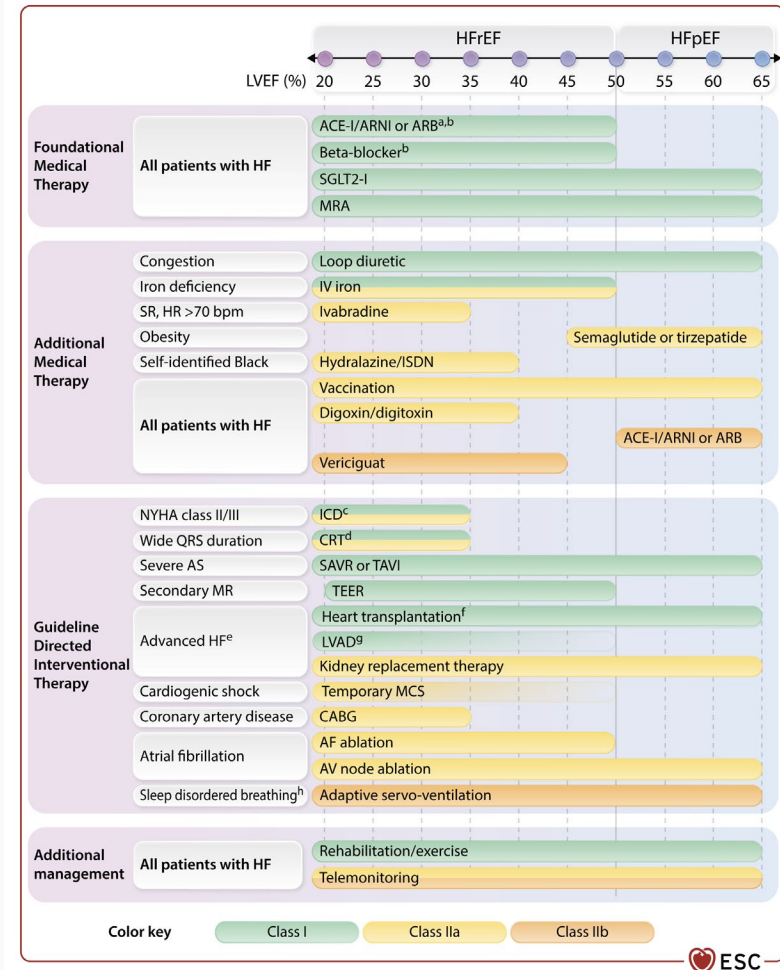
ESC Levels of evidence: therapy and prevention

Level of evidence	Definition
A	Conclusive evidence, usually from at least two adequately powered randomized controlled trials ^a free of major bias, with substantial evidence against the play of chance when combined in a meta-analysis (e.g., $P < 0.005$ for superiority). ^b
B1	Suggestive evidence from at least one adequately powered randomized controlled trial free of major bias, or a meta-analysis of such randomized controlled trials, with some evidence against the play of chance (e.g., $P < 0.05$ for superiority). ^c
B2	Limited evidence from at least two adequately powered non-randomized studies with careful control of major sources of bias and substantial evidence against the play of chance (e.g., $P < 0.005$ for superiority), ^b or from a meta-analysis of small, underpowered randomized controlled trials with some evidence against the play of chance (e.g., $P < 0.05$ for superiority). ^c
C	Preliminary evidence from non-randomized studies without careful control of major sources of bias, a single small, underpowered randomized controlled trial, or expert consensus.

ESC Levels of evidence: diagnostic tests and predictions models

Level of evidence	Diagnostic test	Prediction model (including diagnostic and prognostic models)
A	Conclusive evidence of adequate diagnostic ability from at least two high-quality studies	Conclusive evidence of adequate predictive ability from at least one high-quality derivation study, and two or more external validation studies of at least moderate quality.
B	Suggestive evidence of adequate diagnostic ability from a single high-quality or at least two moderate-quality studies.	Suggestive evidence of adequate predictive ability from one or more derivation and one or more external validation studies of at least moderate quality.
C	Preliminary evidence of adequate diagnostic ability from a single moderate-quality study, any number of low-quality studies, or expert consensus.	Preliminary evidence of adequate predictive ability from a derivation study of at least moderate quality, but with external validation absent or in a low-quality study, or a derivation study of low quality (irrespective of external validation).

Figure 1 Central illustration. Management of heart failure



New concepts

New concepts

HF_rEF has been expanded to include LVEF up to 50%

HF_{mr}EF has been removed

HF_pEF includes LVEF from 50% and above

Decompensated HF replaces acute HF

Stages of HF (A–D) ranging from risk factors to advanced HF have been adopted

New classifications for medical and interventional therapy have been introduced

New recommendations (1)

Recommendations	Class	Level
Prevention of heart failure – Section 4		
Counselling on healthy lifestyle choices including maintaining a healthy weight, consuming a well-balanced diet, avoiding sedentary behaviour, abstaining from smoking and heavy alcohol consumption as well as counselling against the use of drugs such as cocaine, amphetamines and anabolic steroids, is recommended for all patients with stage A or B HF to reduce the risk of HF progression.	I	C
A GLP-1 RA should be considered in patients with T2DM and at least one other additional CV risk factor to reduce the risk of HF or CV death.	IIa	A
<i>Stage B heart failure and left ventricular dysfunction</i>		
An ACE-I or ARB (if intolerant to ACE-I) is recommended in patients with stage B HF and LVEF $\leq 40\%$ to reduce the risk of HFH and death.	I	A
A beta-blocker is recommended in patients with stage B HF and LVEF $< 50\%$ to reduce the risk of HFH or death.	I	B1
Eplerenone is recommended in patients after myocardial infarction with LVEF $\leq 40\%$ with signs of HF or diabetes to reduce the risk of HFH or death.	I	B1

New recommendations (2)

Recommendations	Class	Level
Diagnosis of chronic heart failure – Section 5		
Initial diagnostic testing with serum and urine immunofixation, serum free light chains assay, and DPD/PYP/HMDP bone scintigraphy is recommended in patients with HF and a suspicion of cardiac amyloidosis.	I	B
Genetic testing is recommended in patients fulfilling diagnostic criteria for cardiomyopathy in cases where it enables diagnosis, prognostication, therapeutic stratification, or reproductive management of the patient, or where it enables cascade genetic evaluation of their relatives who would otherwise be enrolled into long-term surveillance.	I	C
18F-FDG-PET scanning should be considered for the diagnostic work-up in patients with cardiomyopathy in whom cardiac sarcoidosis is suspected.	IIa	C
Review of the proportion of RV pacing should be considered in all patients with CIED and de novo or worsening HF _r EF.	IIa	C
Electrical cardioversion as a diagnostic tool may be considered in patients with persistent AF where there is uncertainty about the value of sinus rhythm restoration on symptoms or to assess improvement in left ventricular function.	IIb	C

New recommendations (3)

Recommendations	Class	Level
Management of chronic heart failure – Section 6		
Uptitration of FMT at least every 1–2 weeks in patients with HF, guided by symptoms, vital signs, and laboratory findings, to achieve optimal target doses shown to be efficacious in RCTs is recommended to reduce the risk of HFH or death.	I	C
Continuation of FMT at the highest tolerated doses is recommended in all patients with HF, including those who become asymptomatic or with LVEF improvement, in order to reduce the risk of HFH and death.	I	C
Gradual discontinuation of FMT under frequent clinical, laboratory, and imaging surveillance, may be considered in highly selected patients who are asymptomatic with complete normalization of LV function and volume as well as natriuretic peptide levels after specific treatment of reversible causes of HF, in order to accommodate patient preference.	IIb	C
ACE-I/ARB/ARNI may be considered in patients with symptomatic HFpEF to reduce the risk of HFH.	IIb	C
Initiation of FMT and planning for CRT implantation may be considered simultaneously in patients with symptomatic HFrEF, an LBBB with QRS ≥ 150 ms, and LVEF $\leq 35\%$, to improve symptoms and reduce morbidity and death, although reassessment of LVEF should be conducted prior to CRT implantation.	IIb	C

New recommendations (4)

Recommendations	Class	Level
Decompensated heart failure – Section 7		
<i>Diuretics</i>		
Urinary Na ⁺ -guided diuretic therapy may be considered during the first days of treatment of patients with DHF to improve natriuresis and diuresis.	IIb	B1
<i>SGLT2-I</i>		
In-hospital initiation of SGLT2-I is recommended in patients with DHF after initial stabilization to improve QoL and congestion symptoms and reduce the risk of HFH.	I	B1

New recommendations (5)

Recommendations	Class	Level
Decompensated heart failure – Section 7		
<i>Use of temporary mechanical circulatory support</i>		
A multidisciplinary Shock Team is recommended in potential candidates for temporary MCS to guide the device selection (modality and type) based on patient and HF characteristics.	I	C
Temporary MCS with a microaxial flow pump should be considered in selected patients with cardiogenic shock caused by ST-elevation MI with LV systolic dysfunction and no risk of hypoxic brain injury in order to reduce the risk of death.	IIa	B1
Temporary MCS should be considered in patients with mechanical complications related to MI as a bridge to definitive treatment.	IIa	C
Temporary MCS is not recommended in unselected patients with cardiogenic shock caused by acute MI, due to risk of harm.	III	B1

New recommendations (6)

Recommendations	Class	Level
Advanced heart failure – Section 8		
Early consultation with an advanced HF centre is recommended in patients with advanced HF or at risk of advanced HF, who are motivated and do not have absolute contraindications for heart transplantation or durable MCS, in order to evaluate candidacy.	I	A
Downtitration or discontinuation of beta-blockers or ivabradine should be considered in selected patients with advanced HFrEF and evidence of organ hypoperfusion, despite initial treatment, to increase cardiac output and improve symptoms.	IIa	C
Histopathological examination of explanted heart tissue from LVAD or heart transplantation surgery should be considered in patients with non-ischaemic HF to identify an aetiological diagnosis that may be treatable or facilitate family screening.	IIa	C
Cardiovascular comorbidities – Section 9		
Atrioventricular node ablation combined with CRT should be considered in patients with severely symptomatic permanent AF and poor rate control despite medical therapy and at least one HFH to reduce symptoms, physical limitations, recurrent HFH and death.	IIa	B1
Intravenous amiodarone or digoxin may be considered in haemodynamically unstable patients with HFrEF and AF to stabilize the patient and achieve acute control of heart rate.	IIb	C

New recommendations (7)

Recommendations	Class	Level
Non-cardiovascular comorbidities – Section 10		
Semaglutide or tirzepatide should be considered for patients with symptomatic HF, LVEF $\geq 45\%$, and BMI ≥ 30 kg/m ² , regardless of diabetes status, to reduce body weight, and improve exercise capacity and QoL.	IIa	B1
Bariatric surgery may be considered in obese patients with HF and a BMI ≥ 35 kg/m ² to reduce body weight when repetitive and structured lifestyle changes combined with weight-reducing medications do not result in maintained weight loss.	IIb	C
Adaptive servo-ventilation may be considered in patients with HFrEF and sleep-disordered breathing with predominant obstructive sleep apnoea to improve sleep quality, health-related QoL, and symptoms.	IIb	C
<i>Anxiety, depression, and frailty</i>		
Assessment of anxiety, depression and frailty should be considered in patients with HF to support the development of personalized care plans and to identify factors that may contribute to adverse outcomes.	IIa	C

New recommendations (8)

Recommendations	Class	Level
Multidisciplinary management of heart failure – Section 11		
Personalized exercise training, in the context of multidisciplinary exercise-based cardiac rehabilitation, is recommended for all stable patients, unless there are specific contraindications, in order to improve exercise capacity, and QoL, and reduce the risk of all-cause hospitalization.	I	B1
Personalized exercise training outside exercise-based cardiac rehabilitation is recommended for all stable patients on a long-term basis, unless there are specific contraindications, in order to improve exercise capacity and QoL, and reduce the risk of all-cause hospitalization.	I	B2
Cardiac rehabilitation is recommended in patients with an LVAD to improve functional capacity and QoL.	I	B1
Cardiac rehabilitation should be considered in patients with DHF, as soon as safely possible, to improve functional capacity and QoL.	IIa	B1
<i>Long-term follow-up</i>		
Proactive discussion regarding HF trajectory, goals of care, and advance care planning is recommended in patients with advanced HF to facilitate communication on end of life and QoL.	I	B1
Access to an integrated HF palliative care multidisciplinary team is recommended for patients in an advanced HF stage to improve QoL and reduce symptom burden.	I	B2

New recommendations (9)

Recommendations	Class	Level
Specific conditions – Section 12		
Pre-conception care and counselling is recommended for patients with HF to facilitate decision-making surrounding HF treatments, pregnancy, contraception, pre-implantation genetic screening, and assisted reproductive therapies.	I	C
In pregnant women with HFrEF, it is recommended that non-selective beta-blockers are switched to beta1-selective blockers (metoprolol, bisoprolol) with close mother and foetus monitoring to reduce the risk of adverse foetal events and a lower birth weight.	I	C
ACE-Is, ARBs, ARNIs, MRAs, ivabradine, and SGLT2-Is are not recommended during pregnancy due to the risk of foetotoxicity or teratogenicity.	III	C

Revised recommendations (1)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Prevention of heart failure – Section 4					
Treatment of hypertension is recommended to prevent or delay the onset of HF, and to prevent HF hospitalizations.	I	A	Strict blood pressure control (systolic blood pressure target <130 mmHg) is recommended in patients with hypertension or stage B HF to reduce the risk of HF.	I	A
Diagnosis of chronic heart failure – Section 5					
CMR is recommended for the assessment of myocardial structure and function in those with poor echocardiographic acoustic windows.	I	C	Contrast-enhanced CMR is recommended in patients with suspected cardiomyopathy, or where the underlying aetiology of HF is uncertain if further characterization is likely to add value to patient care.	I	C
CMR is recommended for the characterization of myocardial tissue in suspected infiltrative disease, Fabry disease, inflammatory disease (myocarditis), LV non-compaction, amyloid, sarcoidosis, iron overload/haemochromatosis.	I	C	Contrast-enhanced CMR is recommended in patients with suspected cardiomyopathy, or where the underlying aetiology of HF is uncertain if further characterization is likely to add value to patient care.	I	C

Revised recommendations (2)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
CMR with LGE should be considered in DCM to distinguish between ischaemic and non-ischaemic myocardial damage.	IIa	C	Contrast-enhanced CMR is recommended in patients with suspected cardiomyopathy, or where the underlying aetiology of HF is uncertain if further characterization is likely to add value to patient care.	I	C
Invasive coronary angiography is recommended in patients with angina despite pharmacological therapy or symptomatic ventricular arrhythmias.	I	B	Invasive coronary angiography is recommended in patients with HF, angina, and high probability of obstructive CAD who are potential candidates for revascularization.	I	C
Invasive coronary angiography may be considered in patients with HFrEF with an intermediate to high pre-test probability of CAD and the presence of ischaemia in non-invasive stress tests.	IIb	B	Invasive coronary angiography should be considered in patients with HFrEF and high pre-test likelihood of obstructive CAD, taking into account the risk-to-benefit ratio of the procedures and potential revascularization strategies.	IIa	C

Revised recommendations (3)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Management of chronic heart failure – Section 6					
An MRA is recommended for patients with HFrEF to reduce the risk of HF hospitalization and death.	I	A	An MRA (sMRA for HFrEF and sMRA/nsMRA for HFpEF) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death.	I	A
An MRA may be considered for patients with HFmrEF to reduce the risk of HF hospitalization and death.	IIb	C	An MRA (sMRA for HFrEF and sMRA/nsMRA for HFpEF) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death.	I	A
Digoxin may be considered in patients with symptomatic HFrEF in sinus rhythm despite treatment with an ACE-I (or ARNI), a beta-blocker and an MRA, to reduce the risk of hospitalization (both all-cause and HF hospitalizations).	IIb	B	A cardiac glycoside (digoxin or digitoxin) should be considered in patients with symptomatic HFrEF with LVEF ≤40%, despite optimal FMT, to reduce the risk of HFH.	IIa	B1

Revised recommendations (4)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Management of chronic heart failure – Section 6					
Vericiguat may be considered in patients in NYHA class II–IV who have had worsening HF despite treatment with an ACE-I (or ARNI), a beta-blocker and an MRA to reduce the risk of CV mortality and HF hospitalization. [In patients with HFrEF according to the 2021 classification].	IIb	B	Vericiguat may be considered in patients with symptomatic HFrEF and LVEF <45% despite treatment with optimal FMT to reduce the risk of HFH and CV death.	IIb	B1
CRT rather than RV pacing is recommended for patients with HFrEF regardless of NYHA class or QRS width who have an indication for ventricular pacing for high degree AV block in order to reduce morbidity. This includes patients with AF.	I	A	CRT, rather than RV pacing, should be considered in patients with HFrEF regardless of NYHA class or QRS width who have an indication for ventricular pacing for high-degree AV block in order to reduce the risk of HFH and death.	IIa	B1

Revised recommendations (5)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Decompensated heart failure – Section 7					
Combination of a loop diuretic with thiazide-type diuretic should be considered in patients with resistant oedema who do not respond to an increase in loop diuretic doses.	IIa	B	The addition of short-term intravenous acetazolamide or oral hydrochlorothiazide should be considered in patients with fluid overload previously treated with loop diuretics, to reduce congestion.	IIa	B1
Short-term MCS should be considered in patients with cardiogenic shock as a BTR, BTD, BTB. Further indications include treatment of the cause of cardiogenic shock or long-term MCS or transplantation.	IIa	C	Temporary MCS should be considered in selected patients with HF-related haemodynamic instability as a BTR, BTD, BTB, BTC, or BTT.	IIa	C
IABP is not routinely recommended in post-MI cardiogenic shock.	III	B	IABP is not recommended in unselected patients with cardiogenic shock, due to the lack of effect.	III	B1

Revised recommendations (6)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Advanced heart failure – Section 8					
Continuous inotropes and/or vasopressors may be considered in patients with low cardiac output and evidence of organ hypoperfusion as bridge to MCS or heart transplantation.	IIb	C	Continuous inotropes should be considered in patients with advanced HFrEF, low cardiac output, and evidence of hypoperfusion or end-organ dysfunction, as BTD, BTT, or as bridge to durable MCS, to increase cardiac output and improve symptoms.	IIa	C
Long-term MCS should be considered in patients with advanced HFrEF despite optimal medical and device therapy, not eligible for heart transplantation or other surgical options, and without severe right ventricular dysfunction, to reduce the risk of death and improve symptoms.	IIa	A	Durable MCS (LVAD) is recommended in selected patients with advanced HFrEF, despite FMT and GDIT, as BTT, BTC, BTR, or as destination therapy to improve symptoms and reduce the risk of death.	I	C

Revised recommendations (7)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Advanced heart failure – Section 8					
Long-term MCS should be considered in patients with advanced HFrEF refractory to optimal medical and device therapy as a bridge to cardiac transplantation in order to improve symptoms, reduce the risk of HF hospitalization and the risk of premature death.	IIa	B	Durable MCS (LVAD) is recommended in selected patients with advanced HFrEF, despite FMT and GDIT, as BTT, BTC, BTD, or as destination therapy to improve symptoms and reduce the risk of death.	I	C

Revised recommendations (8)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Cardiovascular comorbidities – Section 9					
Beta-blockers should be considered for short- and long-term rate control in patients with HF and AF.	Ila	B	Beta-blockers are recommended in stable patients with HFrEF and AF as first-line therapy for short- and long-term rate control.	I	C
In cases of a clear association between paroxysmal or persistent AF and worsening of HF symptoms, which persist despite MT, catheter ablation should be considered for the prevention or treatment of AF.	Ila	B	Catheter ablation for AF should be considered in selected patients with symptomatic AF and HFrEF to improve QoL and reduce the risk of HFH or death.	Ila	C
Coronary revascularization should be considered to relieve persistent symptoms of angina (or an angina-equivalent) in patients with HFrEF, CCS, and coronary anatomy suitable for revascularization, despite OMT including anti-anginal drugs.	Ila	C	Coronary revascularization is recommended in patients with HF, obstructive CAD, and persistent angina, despite FMT for HF and antianginal drugs, to relieve symptoms.	I	C

Revised recommendations (9)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Cardiovascular comorbidities – Section 9					
Aortic valve intervention, TAVI or SAVR, is recommended in patients with HF and severe high-gradient aortic stenosis to reduce mortality and improve symptoms.	I	B	Aortic valve intervention, SAVR or TAVI, is recommended in patients with severe aortic stenosis and HF to improve symptoms and reduce the risk of death.	I	B1
Percutaneous edge-to-edge mitral valve repair should be considered in carefully selected patients with secondary mitral regurgitation, not eligible for surgery and not needing coronary revascularization, who are symptomatic despite OMT and who fulfil criteria for achieving a reduction in HF hospitalizations.	IIa	B	Mitral transcatheter edge-to-edge repair is recommended in haemodynamically stable, symptomatic patients with HFrEF and persistent severe secondary mitral regurgitation despite optimized FMT and CRT if indicated, who fulfil specific clinical and echocardiographic criteria, in order to reduce the risk of HFH and improve QoL.	I	B1

Revised recommendations (10)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Multidisciplinary management – Section 11					
Monitoring of pulmonary artery pressure using a wireless haemodynamic monitoring system may be considered in symptomatic patients with HF in order to improve clinical outcomes.	IIb	B	Remote haemodynamic monitoring of PA pressure should be considered in symptomatic patients with HF, NYHA class III and a HFH in the past 12 months in order to reduce the risk of HFH.	IIa	B1

Revised recommendations (11)

Recommendations in 2021 version	Class	Level	Recommendations in 2026 version	Class	Level
Specific conditions – Section 12					
Tafamidis is recommended in patients with genetic testing proven hTTR-CA and NYHA class I or II symptoms to reduce symptoms, CV hospitalization and mortality.	I	B	Treatment with a TTR silencer (vutrisiran) or stabilizer (tafamidis or acoramidis) for patients with variant or wild-type transthyretin cardiac amyloidosis and NYHA class I–III is recommended to reduce progression of symptoms and the risk of CV hospitalization and all-cause death.	I	A
Tafamidis is recommended in patients with wtTTR-CA and NYHA class I or II symptoms to reduce symptoms, CV hospitalization and mortality.	I	B	Treatment with a TTR silencer (vutrisiran) or stabilizer (tafamidis or acoramidis) for patients with variant or wild-type transthyretin cardiac amyloidosis and NYHA class I–III is recommended to reduce progression of symptoms and the risk of CV hospitalization and all-cause death.	I	A

Figure 2 Heart failure stages and classifications

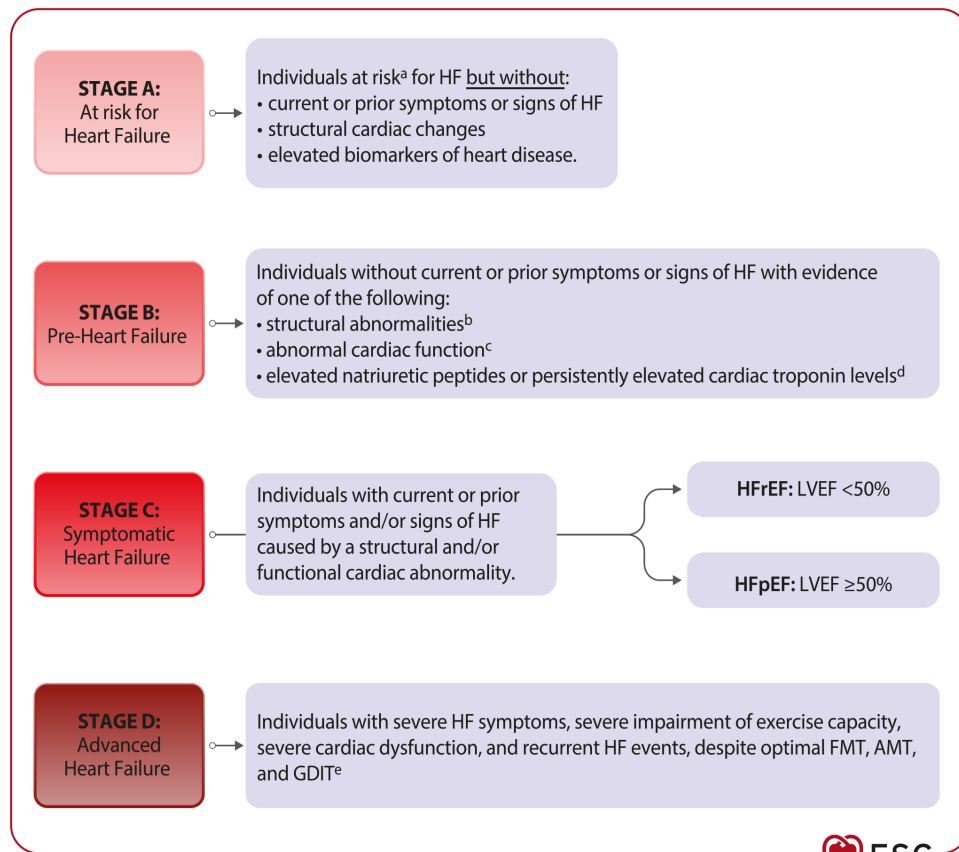
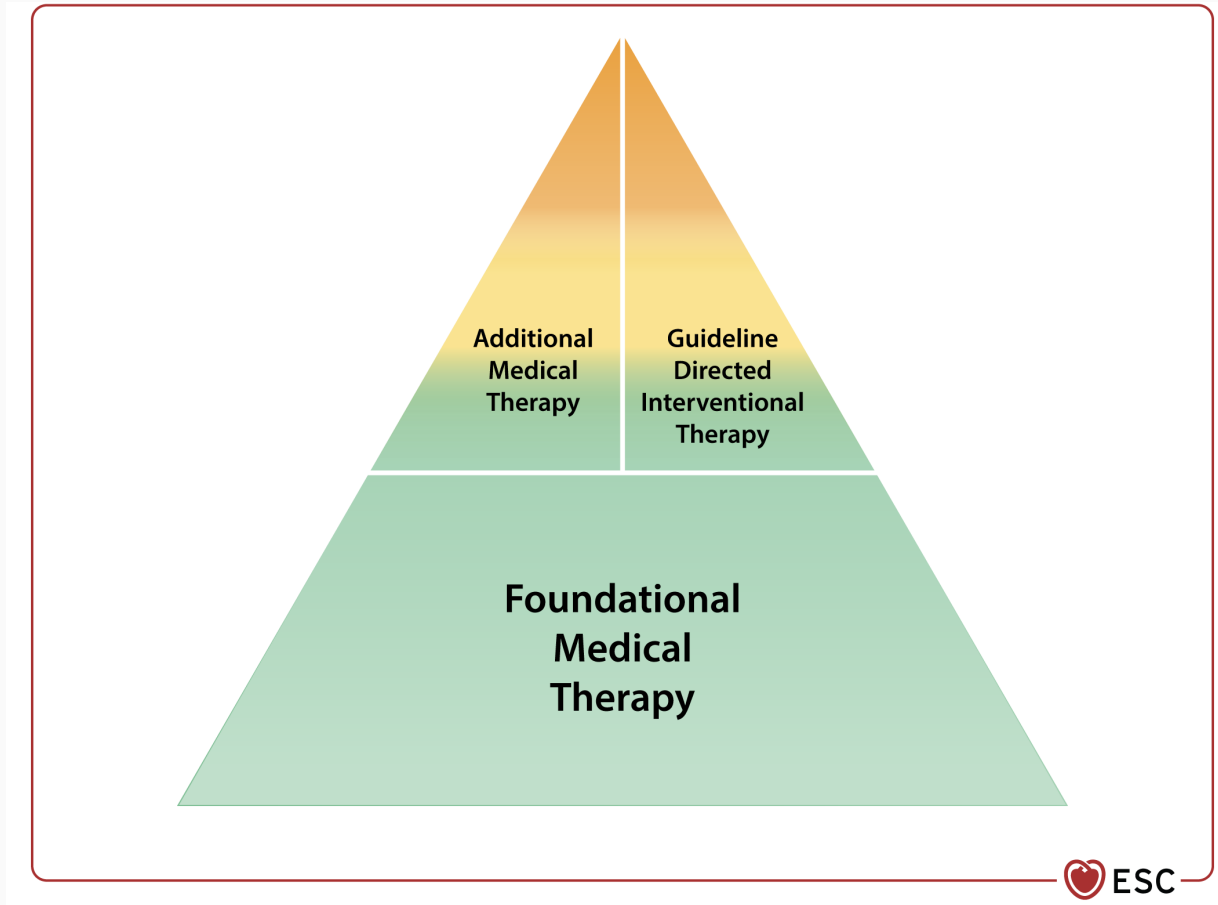


Figure 3 New nomenclature for heart failure therapies



Recommendations for prevention of heart failure

Recommendations	Class	Level
Strict blood pressure control (systolic blood pressure target <130 mmHg) is recommended in patients with hypertension or stage B HF to reduce the risk of HF.	I	A
An SGLT2-I is recommended in patients with T2DM at risk of HF (multiple ASCVD risk factors or established ASCVD) to reduce the risk of HF and CV death.	I	A
An SGLT2-I is recommended in patients with T2DM and CKD to reduce the risk of HF or CV death.	I	A
Finerenone is recommended in patients with T2DM and CKD to reduce the risk of HF.	I	A
Counselling on healthy lifestyle choices including maintaining a healthy weight, consuming a well-balanced diet, avoiding sedentary behaviour, abstaining from smoking and heavy alcohol consumption as well as counselling against the use of drugs such as cocaine, amphetamines, and anabolic steroids is recommended for all patients with stage A or B HF to reduce the risk of HF progression.	I	C
Treatment with a statin should be considered in people at high risk of, or with established ASCVD, to reduce the risk of HF.	IIa	A
A GLP-1 RA should be considered in patients with T2DM and at least one other additional CV risk factor to reduce the risk of HF or CV death.	IIa	A

Recommendations for stage B heart failure and left ventricular dysfunction

Recommendations	Class	Level
An ACE-I or ARB (if intolerant to ACE-I) is recommended in patients with stage B HF and LVEF $\leq 40\%$ to reduce the risk of HFH and death.	I	A
A beta-blocker is recommended in patients with stage B HF and LVEF $< 50\%$ to reduce the risk of HFH or death.	I	B1
Eplerenone is recommended in patients with LVEF $\leq 40\%$ after myocardial infarction with signs of HF or diabetes to reduce the risk of HFH and death.	I	B1

Symptoms of heart failure

<i>Typical</i>	<i>Less typical</i>
• Dyspnoea • Orthopnoea • Paroxysmal nocturnal dyspnoea • Reduced exercise tolerance • Fatigue, tiredness, increased time to recover after exercise • Ankle oedema	• Nocturnal cough • Wheezing • Bloated feeling • Loss of appetite • Confusion (especially in the elderly) • Depression • Palpitations • Dizziness • Syncope • Bendopnoe

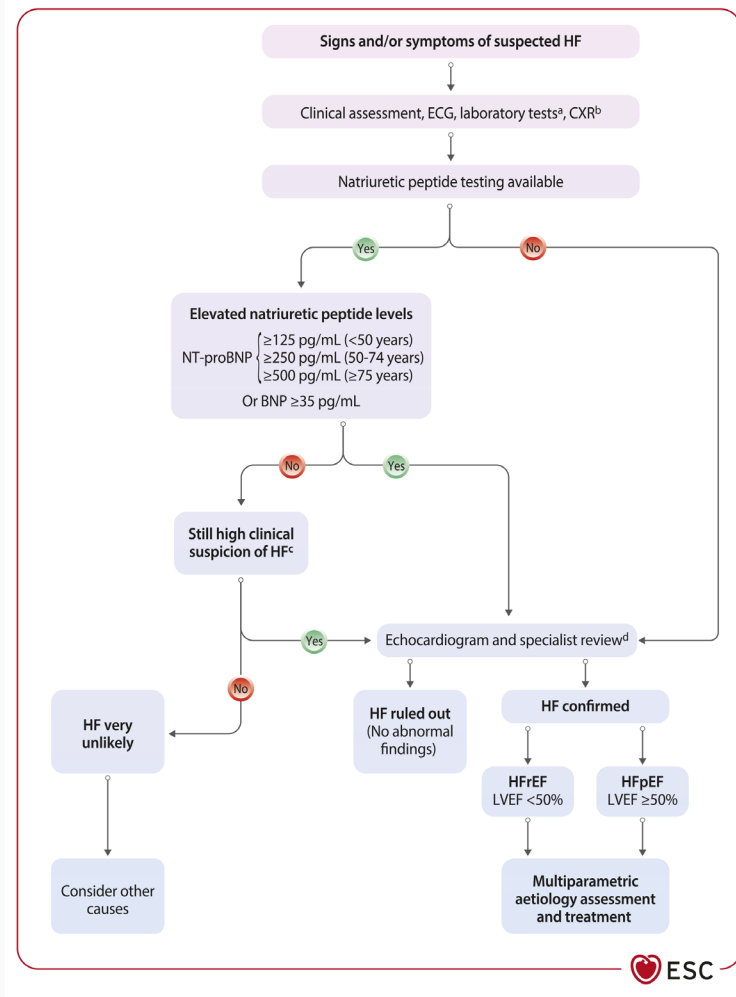
Signs of heart failure

More specific	Less specific
• Elevated jugular venous pressure • Hepatojugular reflux • Third heart sound (gallop rhythm) • Laterally displaced apical impulse	• Weight gain (>2 kg/week) • Weight loss (in advanced HF) • Tissue wasting (cachexia) • Cardiac murmur • Peripheral oedema (ankle, sacral, scrotal) • Pulmonary crepitations • Pleural effusion • Tachycardia • Irregular pulse • Tachypnoea • Cheyne–Stokes respiration • Hepatomegaly • Ascites • Cold extremities • Oliguria • Narrow pulse pressure

New York Heart Association functional classification based on severity of symptoms and physical activity

NYHA class	Severity of symptoms and physical activity
<i>Class I</i>	No limitation of physical activity. Ordinary physical activity does not cause undue breathlessness, fatigue, or palpitations.
<i>Class II</i>	Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in undue breathlessness, fatigue, or palpitations.
<i>Class III</i>	Marked limitation of physical activity. Comfortable at rest, but less than ordinary activity results in undue breathlessness, fatigue, or palpitations.
<i>Class IV</i>	Unable to carry on any physical activity without discomfort. Symptoms at rest can be present. If any physical activity is undertaken, discomfort is increased.

Figure 4 Diagnostic algorithm for chronic heart failure presenting in the outpatient or community setting



Causes for increase or decrease in natriuretic peptide levels (1)

Causes for increased natriuretic peptide levels

- | | |
|---|---|
| <ul style="list-style-type: none">• Acute coronary syndrome• Advanced agea• Atrial fibrillation• Cardiac contusion• COPD• Critical illness• Female sex• Kidney dysfunction• LV hypertrophy• Myocarditis• Obstructive sleep apnoea• Paraneoplastic syndrome | <ul style="list-style-type: none">• Pulmonary embolism• Pulmonary hypertension• RV pacing• Severe anaemia• Severe endocrine dysfunction• Severe infections (including pneumonia and sepsis)• Severe liver disease• Severe metabolic acidosis and diabetic ketoacidosis• Subarachnoid haemorrhage• Systemic hypertension• Valvular heart disease |
|---|---|

Causes for increase or decrease in natriuretic peptide levels (2)

Causes for decreased natriuretic peptide levels

- | | |
|--|---|
| <ul style="list-style-type: none">• African ancestry• ARNIb/ACE-I/ARB• Beta-blockers• Constrictive pericarditis | <ul style="list-style-type: none">• MRA• Obesity• SGLT2-I |
|--|---|

Recommendations for diagnostic investigations in all patients with suspected heart failure

Recommendations	Class	Level
In patients with suspected HF, the following laboratory tests are recommended when screening for comorbidities: full blood count, kidney function (eGFR and UACR), electrolytes, liver function, thyroid function, HbA1c, lipids, and iron status (TSAT and ferritin).	I	C
Natriuretic peptide measurement is recommended in patients with suspected HF (interpreted in relation to patient age, obesity and other factors known to affect natriuretic peptide level).	I	C
A 12-lead ECG is recommended in patients with suspected HF.	I	C
Chest radiography (X-ray) is recommended in patients with suspected HF.	I	C
Transthoracic echocardiogram is recommended in patients with suspected HF to confirm the diagnosis, differentiate between HF phenotypes and aid in identifying the underlying aetiology of HF.	I	C

Simplified echocardiographic criteria for supporting objective evidence of heart failure with preserved ejection fraction

Criteria	Echocardiographic threshold
LV hypertrophy	LV mass index $\geq 95 \text{ g/m}^2$ (female) $\geq 115 \text{ g/m}^2$ (male) OR Relative wall thickness > 0.42
LA dilatation (volume indexed by BSA)	$> 34 \text{ mL/m}^2$ (sinus rhythm) $> 40 \text{ mL/m}^2$ (AF)
Increased likelihood of elevated LV filling pressure	$E/e' > 9$ at rest
Raised estimated systolic PA pressure	$> 35 \text{ mmHg}$ OR TR velocity at rest $> 2.8 \text{ m/s}$

Figure 5 Major contributing causes of heart failure with reduced ejection fraction

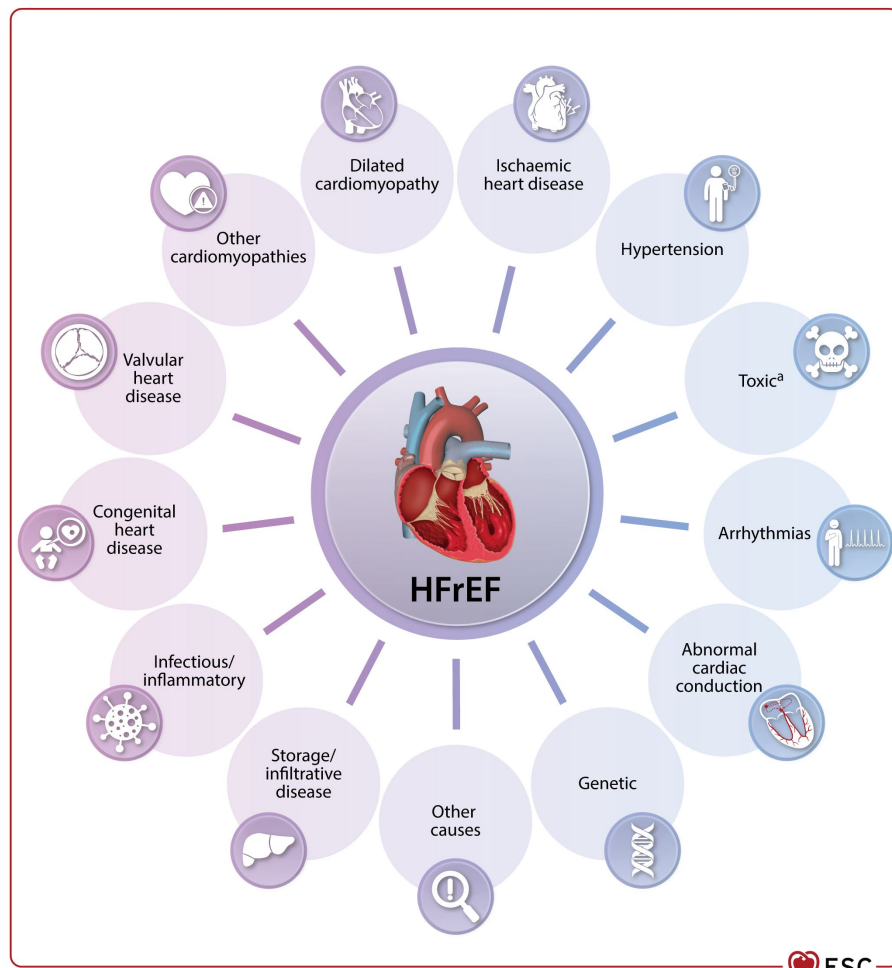


Figure 6 Major risk factors of heart failure with preserved ejection fraction

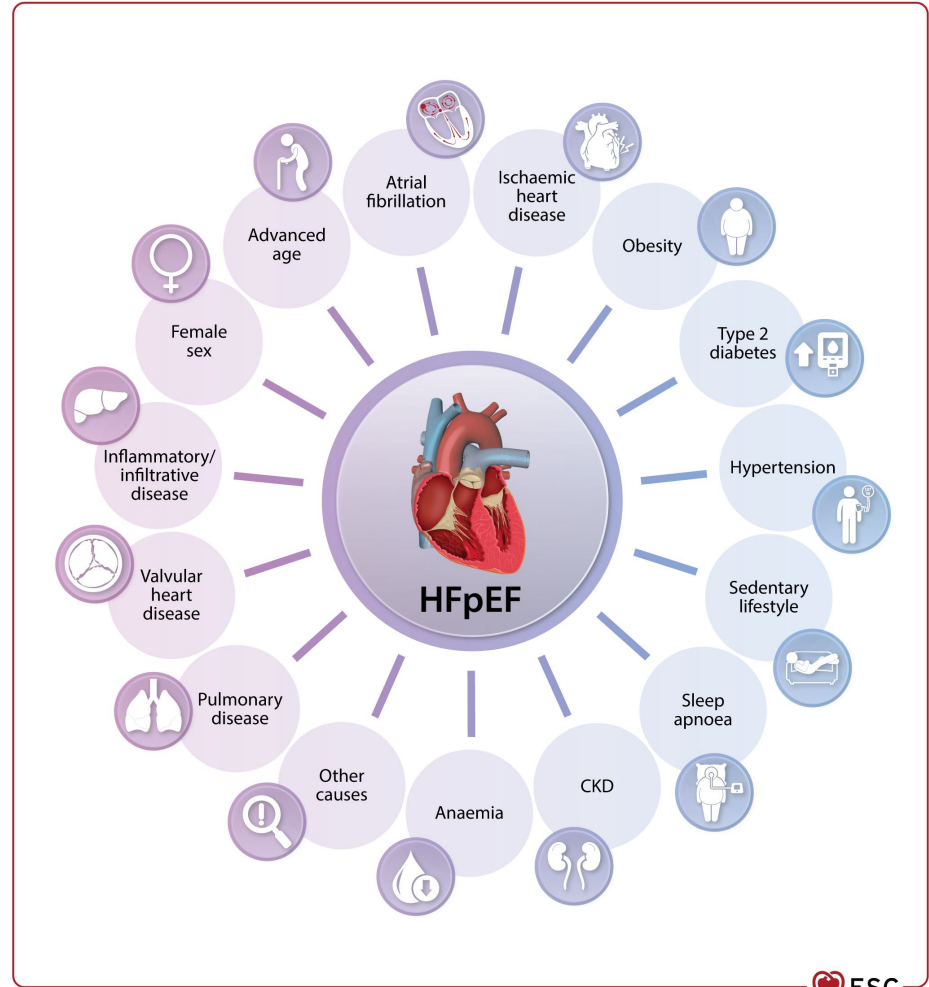


Figure 7

Multiparametric approach to aetiology assessment in heart failure

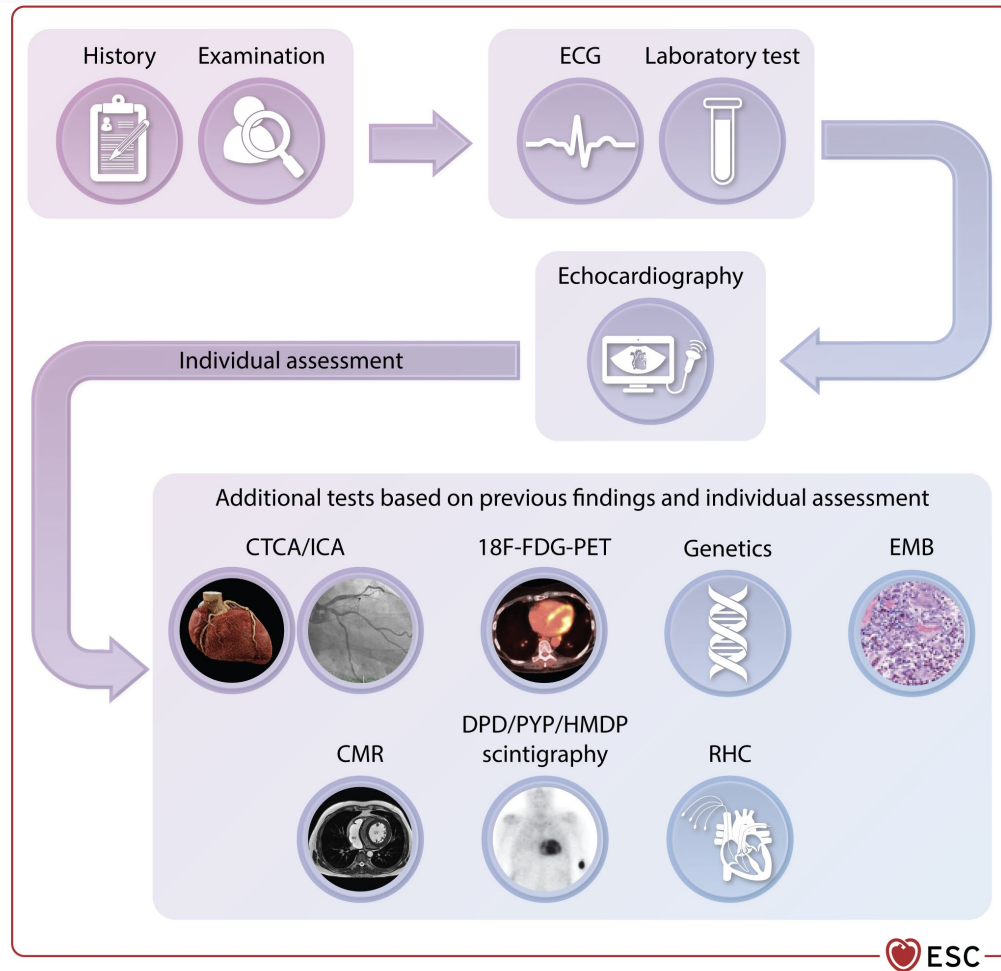


Figure 8 Diagnosis of heart failure due to ischaemic aetiology

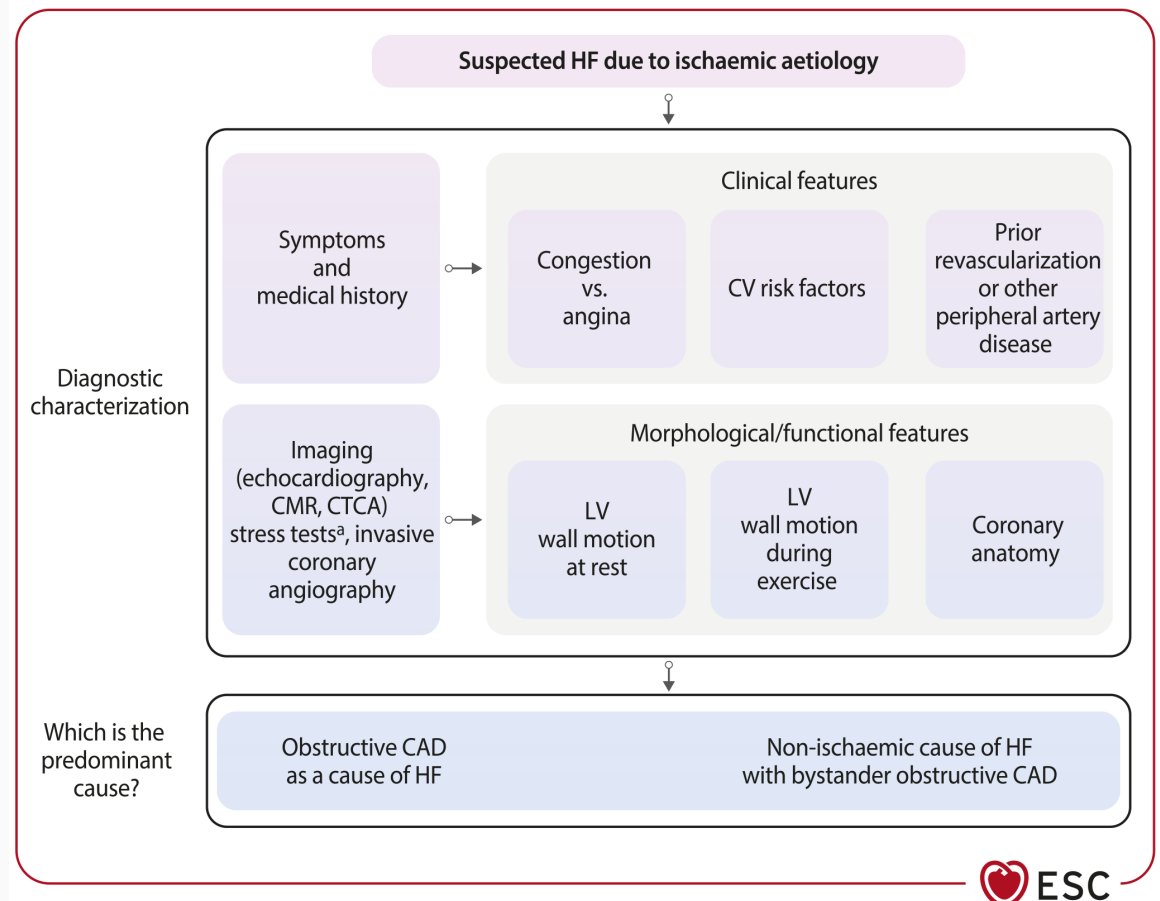
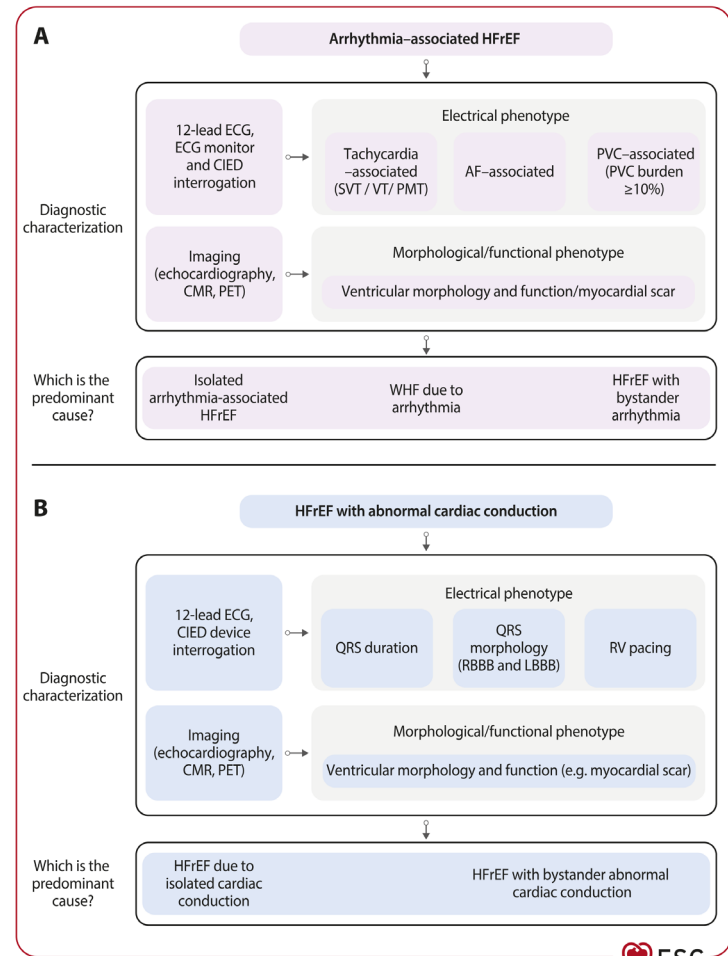


Figure 9 Diagnosing arrhythmia-associated heart failure and heart failure with abnormal cardiac conduction



Recommendations for specialized diagnostic investigations in patients with established heart failure to detect reversible/treatable causes of heart failure (1)

Recommendations	Class	Level
Investigations for underlying aetiology in patients with HFpEF or HFrEF		
Contrast-enhanced CMR is recommended in patients with suspected cardiomyopathy, or where the underlying aetiology of HF is uncertain if further characterization is likely to add value to patient care.	I	C
Initial diagnostic testing with serum and urine immunofixation, serum free light chains assay, and DPD/PYP/HMDP bone scintigraphy is recommended in patients with HF and a suspicion of cardiac amyloidosis.	I	B
Genetic testing is recommended in patients fulfilling diagnostic criteria for cardiomyopathy in cases where it enables diagnosis, prognostication, therapeutic stratification, or reproductive management of the patient, or where it enables cascade genetic evaluation of their relatives who would otherwise be enrolled into long-term surveillance.	I	C
Invasive coronary angiography is recommended in patients with HF, angina, and high probability of obstructive CAD who are potential candidates for revascularization.	I	C

Recommendations for specialized diagnostic investigations in patients with established heart failure to detect reversible/treatable causes of heart failure (2)

Recommendations	Class	Level
Investigations for underlying aetiology in patients with HFpEF or HFrEF		
18F-FDG-PET scanning should be considered for the diagnostic work-up in patients with cardiomyopathy in whom cardiac sarcoidosis is suspected.	IIa	C
Endomyocardial biopsy should be considered to aid in diagnosis and management in patients with rapidly progressive HF despite standard therapy or when the results of other clinical investigations suggest myocardial inflammation, infiltration, or storage that cannot be identified by other means.	IIa	C
Right heart catheterization should be considered in patients where HF is thought to be due to constrictive pericarditis or congenital heart disease.	IIa	C
Right heart catheterization may be considered in selected patients with suspected HFpEF or structural heart disease to confirm the diagnosis, or when clinical examination and non-invasive testing are insufficient to understand the haemodynamic state.	IIb	C

Recommendations for specialized diagnostic investigations in patients with established heart failure to detect reversible/treatable causes of heart failure (3)

Recommendations	Class	Level
Investigations for patients with HFrEF		
Invasive coronary angiography should be considered in patients with HFrEF and high pre-test likelihood of obstructive CAD, taking into account the risk-to-benefit ratio of the procedures and potential revascularization strategies.	IIa	C
CTCA should be considered in patients with HFrEF and low-moderate pre-test likelihood of obstructive CAD.	IIa	C
Review of the proportion of RV pacing should be considered in all patients with CIED and de novo or worsening HFrEF.	IIa	C
Electrical cardioversion as a diagnostic tool may be considered in patients with persistent AF where there is uncertainty about the value of sinus rhythm restoration on symptoms or to assess improvement in left ventricular function.	IIb	C

Evidence-based doses of disease-modifying drugs in key randomized trials in patients with heart failure (1)

Drug	Starting dose	Target dose
ACE-Is		
Captopril	6.25 mg t.i.d.	50 mg t.i.d.
Enalapril	2.5 mg b.i.d.	10-20 mg b.i.d.
Lisinopril	2.5-5 mg o.d.	20-35 mg o.d.
Ramipril	1.25-2.5 mg b.i.d.	5 mg b.i.d.
Trandolapril	0.5 mg o.d.	4 mg o.d.
ARNIs		
Sacubitril–valsartan	49/51 mg b.i.d.	97/103 mg b.i.d.
ARBs		
Candesartan	4 mg o.d.	32 mg o.d.
Losartan	(12.5) 25–50 mg o.d.	150 mg o.d.
Valsartan	40 mg b.i.d.	160 mg b.i.d.
Beta-blockers		
Bisoprolol	1.25 mg o.d.	10 mg o.d.
Carvedilol	3.125 mg b.i.d.	25 mg b.i.d.d
Metoprolol succinate (CR/XL)	12.5-25 mg o.d.	200 mg o.d.
Nebivolol	1.25 mg o.d.	10 mg o.d.

Evidence-based doses of disease-modifying drugs in key randomized trials in patients with heart failure (2)

Drug	Starting dose	Target dose
MRAs		
<i>Steroidal</i>		
Eplerenone	25 mg o.d.	50 mg o.d.
Spironolactone	12.5-25 mg o.d.	50 mg o.d.
<i>Non-steroidal</i>		
Finerenone	10–20 mg o.d.	20–40 mg o.d.
SGLT2-Is		
Dapagliflozin	10 mg o.d.	10 mg o.d.
Empagliflozin	10 mg o.d.	10 mg o.d.
Other agents		
Digoxin	62.5 µg o.d.	250 µg o.d.
Digitoxin	0.07 mg o.d.	0.1 mg o.d.
Ivabradine	5 mg b.i.d.	7.5 mg b.i.d.
Vericiguat	2.5 mg o.d.	10 mg o.d.
Hydralazine/isosorbide dinitrate	37.5 mg t.i.d./20 mg t.i.d.	75 mg t.i.d./40 mg t.i.d.

Recommendations for pharmacological management of chronic heart failure (1)

Recommendations	Class	Level
All patients with HF, independent of LVEF		
A SGLT2-I (dapagliflozin or empagliflozin) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death.	I	A
An MRA (sMRA for HFrEF and sMRA/nsMRA for HFpEF) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death.	I	A
Dynamic individualized dosing of loop diuretics according to volume status is recommended in patients with HF and signs and/or symptoms of congestion to alleviate HF symptoms, improve exercise capacity, and reduce the risk of HFH.	I	A
Uptitration of FMT at least every 1–2 weeks in patients with HF, guided by symptoms, vital signs, and laboratory findings, to achieve optimal target doses shown to be efficacious in RCTs is recommended to reduce the risk of HFH or death.	I	C
Continuation of FMT at the highest tolerated doses is recommended in all patients with HF, including those who become asymptomatic or with LVEF improvement, in order to reduce the risk of HFH and death.	I	C

Recommendations for pharmacological management of chronic heart failure (2)

Recommendations	Class	Level
HFrEF		
A beta-blocker is recommended in stable patients with symptomatic HFrEF to reduce the risk of HFH and death.	I	A
An ACE-I or ARNI is recommended in patients with symptomatic HFrEF to reduce the risk of HFH and death.	I	A
Switching from an ACE-I or ARB to an ARNI is recommended in patients with symptomatic HFrEF to reduce the risk of HFH and death.	I	B1
An ARB is recommended in patients with symptomatic HFrEF to reduce the risk of HFH and CV death if they are unable to tolerate an ACE-I or ARNI.	I	A
A cardiac glycoside (digoxin or digitoxin) should be considered in patients with symptomatic HFrEF with LVEF $\leq 40\%$, despite optimal FMT, to reduce the risk of HFH.	IIa	B1
Ivabradine should be considered for patients with symptomatic HFrEF, LVEF $\leq 35\%$, in sinus rhythm, resting heart rate >70 b.p.m., on the highest tolerated ACE-I/ARNI, MRA, SGLT2-I and beta-blockers to reduce the risk of HFH or CV death.	IIa	B1

Recommendations for pharmacological management of chronic heart failure (3)

Recommendations	Class	Level
HFrEF		
Ivabradine should be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, in sinus rhythm, who are unable to tolerate a beta-blocker, on the highest tolerated ACE-I/ARNI and MRA to reduce the risk of HFH or CV death.	IIa	C
Hydralazine and isosorbide dinitrate should be considered in self-identified Black patients with symptomatic HFrEF with LVEF $\leq 40\%$ on top of optimal FMT to reduce the risk of HFH and death.	IIa	B2
Hydralazine and isosorbide dinitrate may be considered in patients with symptomatic HFrEF with LVEF $\leq 40\%$ who cannot tolerate ARNI, ACE-I, or ARB to reduce the risk of HFH and death.	IIb	C
Vericiguat may be considered in patients with symptomatic HFrEF and LVEF $< 45\%$ despite treatment with optimal FMT to reduce the risk of HFH and CV death.	IIb	B1
Gradual discontinuation of FMT under frequent clinical, laboratory, and imaging surveillance, may be considered in highly selected patients who are asymptomatic with complete normalization of LV function and volume as well as natriuretic peptide levels after specific treatment of reversible causes of HF, in order to accommodate patient preference.	IIb	C
HFpEF		
ACE-I/ARB/ARNI may be considered in patients with symptomatic HFpEF to reduce the risk of HFH.	IIb	C

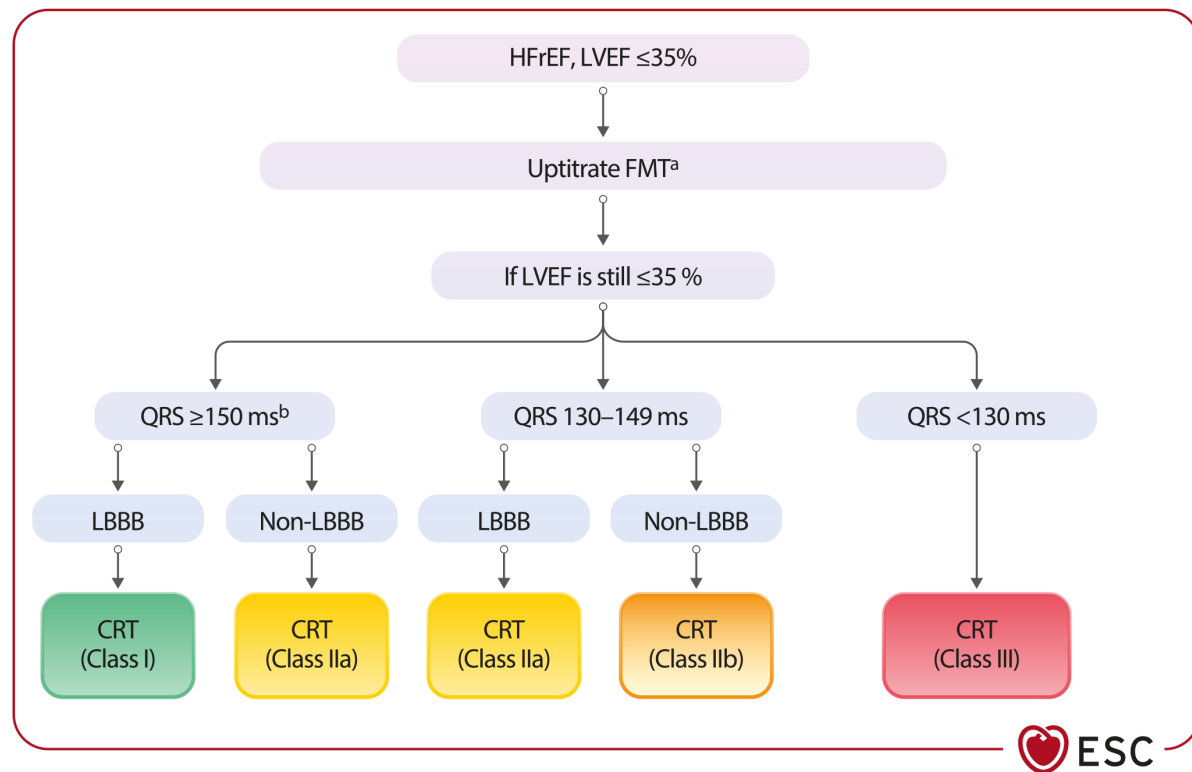
Recommendations for implantable cardioverter-defibrillator implantation (1)

Recommendations	Class	Level
An ICD is recommended in patients who have recovered from a ventricular arrhythmia causing haemodynamic instability (secondary prevention), and who are expected to survive for >1 year with good functional status, in the absence of reversible causes, or unless the ventricular arrhythmia has occurred <48 h after an MI, to reduce the risk of sudden death and all-cause death.	I	A
An ICD is recommended in patients with symptomatic HFrEF (NYHA class II/III) of an ischaemic aetiology (unless they have had an MI in the prior 40 days), and an LVEF $\leq 35\%$ despite ≥ 3 months of optimal FMT, provided they are expected to survive longer than 1 year with good functional status, to reduce the risk of sudden death and all-cause death.	I	B1
An ICD should be considered in patients with symptomatic HFrEF (NYHA class II/III) of a non-ischaemic aetiology, and an LVEF $\leq 35\%$ despite ≥ 3 months of optimal FMT, provided they are expected to survive longer than 1 year with good functional status, to reduce the risk of sudden death and all-cause death.	IIa	B1
Prior to generator replacement for CIEDs, a shared decision-making process between the patient, and a cardiologist experienced in CIEDs and HF should be considered to ensure that the balance between risks and benefits still favour a CIED.	IIa	C

Recommendations for implantable cardioverter-defibrillator implantation (2)

Recommendations	Class	Level
A wearable cardioverter-defibrillator may be considered in patients with HF who are at risk of sudden cardiac death for a limited period in order to increase survival as a bridge to decision for permanent ICD implantation or while listed for heart transplantation.	IIb	C
ICD implantation for primary prevention is not recommended within 40 days of an MI as implantation during this period does not improve prognosis.	III	B1
ICD implantation is not recommended in patients in NYHA class IV with severe symptoms refractory to pharmacological therapy unless they are candidates for CRT, a ventricular assist device, or heart transplantation.	III	C

Figure 10 Indications for cardiac resynchronization therapy implantation in heart failure



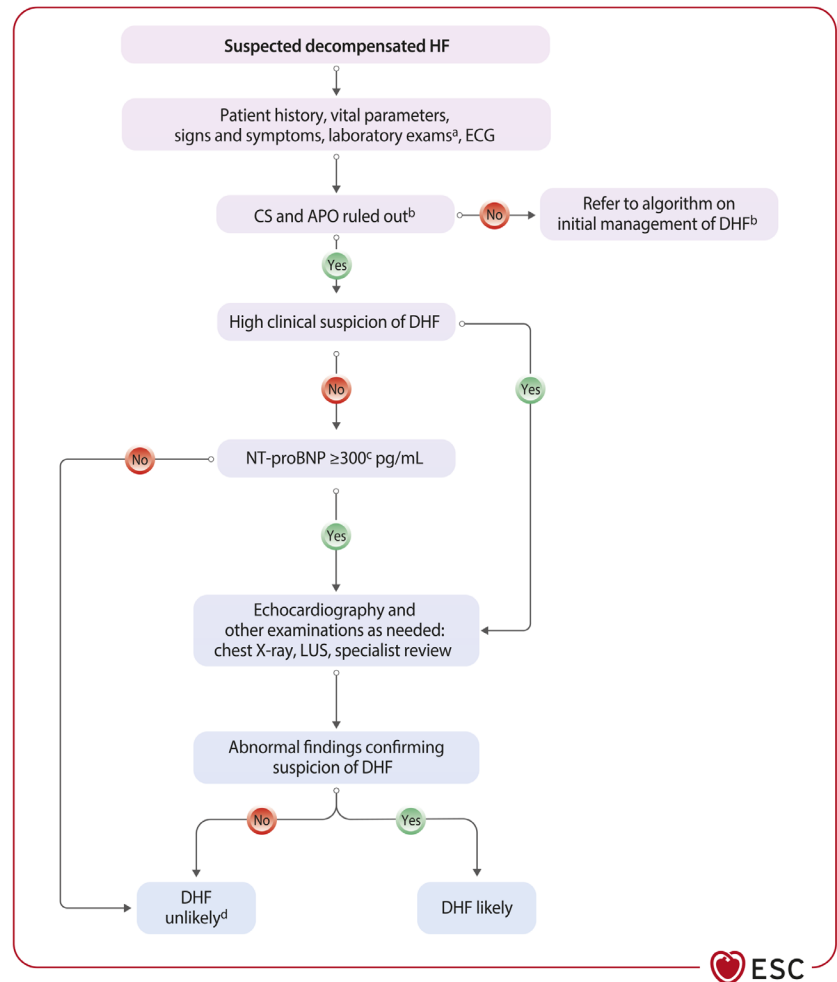
Recommendations for cardiac resynchronization therapy implantation (1)

Recommendations	Class	Level
CRT is recommended in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, in SR with LBBB and QRS duration ≥ 150 ms in order to improve symptoms and reduce the risk of hospitalizations and death.	I	A
An upgrade to CRT should be considered in patients with an LVEF $\leq 35\%$ who have received a conventional pacemaker or an ICD and subsequently develop worsening HF despite optimal FMT and who have a significant proportion of RV pacing in order to reduce the risk of HFH or death.	IIa	B1
CRT, rather than RV pacing, should be considered in patients with HFrEF regardless of NYHA class or QRS width who have an indication for ventricular pacing for high-degree AV block in order to reduce the risk of HFH and death.	IIa	B1
CRT should be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, who are in SR with non-LBBB QRS morphology and QRS duration ≥ 150 ms in order to improve symptoms and reduce morbidity and death.	IIa	C
CRT should be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, who are in SR with LBBB QRS morphology and QRS duration of 130–149 ms in order to improve symptoms and reduce morbidity and death.	IIa	C

Recommendations for cardiac resynchronization therapy implantation (2)

Recommendations	Class	Level
CRT may be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, who are in SR with non-LBBB QRS morphology and QRS duration of 130–149 ms in order to improve symptoms and reduce morbidity and death.	IIb	C
Initiation of FMT and planning for CRT implantation may be considered simultaneously in patients with symptomatic HFrEF, an LBBB with QRS ≥ 150 ms, and LVEF $\leq 35\%$, to improve symptoms and reduce morbidity and death, although reassessment of LVEF should be conducted prior to CRT implantation.	IIb	C
CRT is not recommended in patients with a QRS duration < 130 ms who do not have an indication for pacing due to high-degree AV block.	III	A

Figure 11 Diagnosing decompensated heart failure



Signs, symptoms, and altered laboratory tests in congestion and hypoperfusion profiles

Left-sided congestion	Right-sided congestion	Hypoperfusion
• Dyspnoea • Orthopnoea • Cough • Tachypnoea • Rales • S3 • Pleural effusion • Elevated natriuretic peptides	• Peripheral oedema • Abdominal distension • Hepatomegaly • Jugular vein distension • Hepatojugular reflux • Pleural effusion • Elevated serum creatinine • Elevated bilirubin, ALP, GGT • Slightly elevated natriuretic peptides	• Cold, sweaty extremities • Pale skin • Dizziness • Mental confusion • Oliguria • Narrow pulse pressure • Elevated serum lactate • Elevated serum creatinine • Elevated aminotransferase

SCAI classification for cardiogenic shock

STAGE A 'At risk'	A patient who is not currently experiencing signs or symptoms of CS but is at risk for development of CS. These patients may include those with acute MI and acute or chronic heart failure.
STAGE B 'Beginning/Pre-shock' cardiogenic shock	A patient who has clinical evidence of hemodynamic instability (including relative hypotension or tachycardia) without hypoperfusion.
STAGE C 'Classic' cardiogenic shock	A patient that manifests with hypoperfusion requiring intervention (inotrope, vasopressor, or mechanical support) beyond volume resuscitation to restore perfusion. These patients typically present with relative hypotension.
STAGE D 'Deteriorating'	A patient that is similar to stage C but deteriorating with failure to respond to initial interventions.
STAGE E 'Extremis'	A patient with refractory shock or cardiac arrest requiring multiple simultaneous acute interventions, including CPR.

Figure 12 Phases and goals for in-hospital management of decompensated heart failure

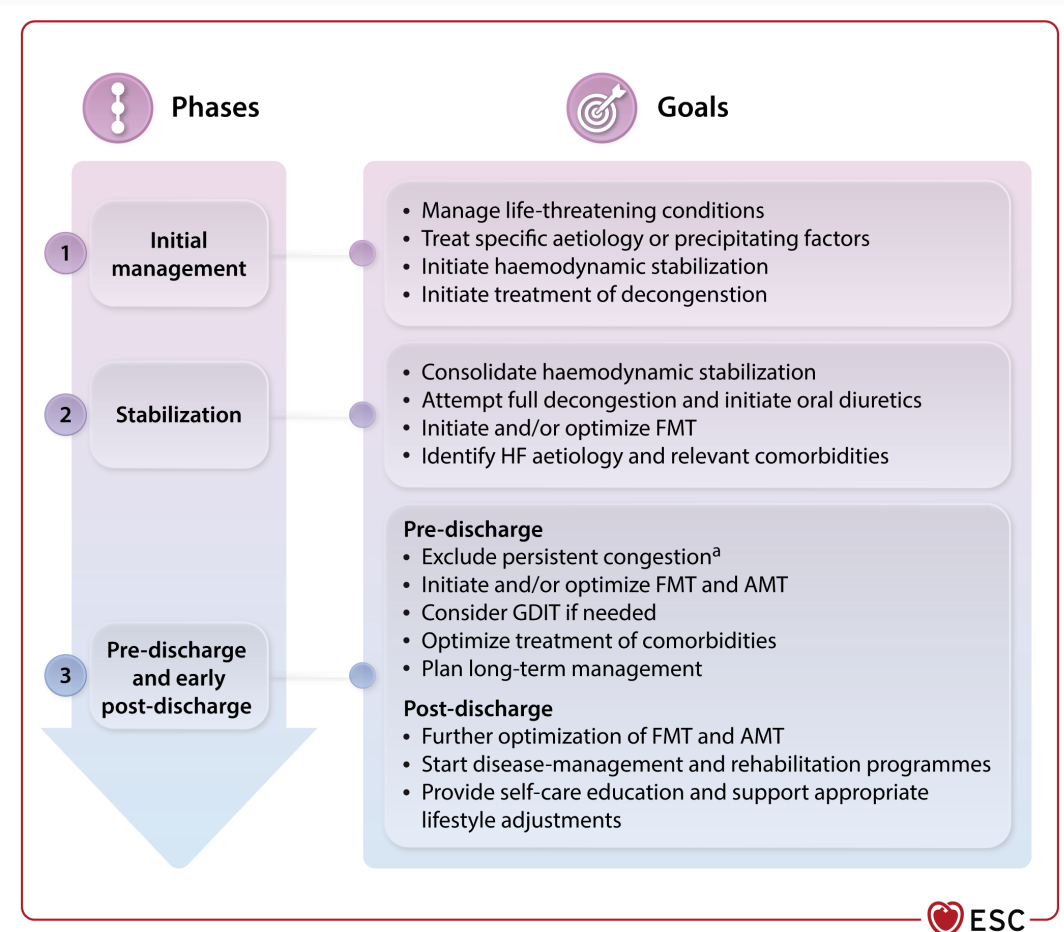


Figure 13

Management of decompensated heart failure

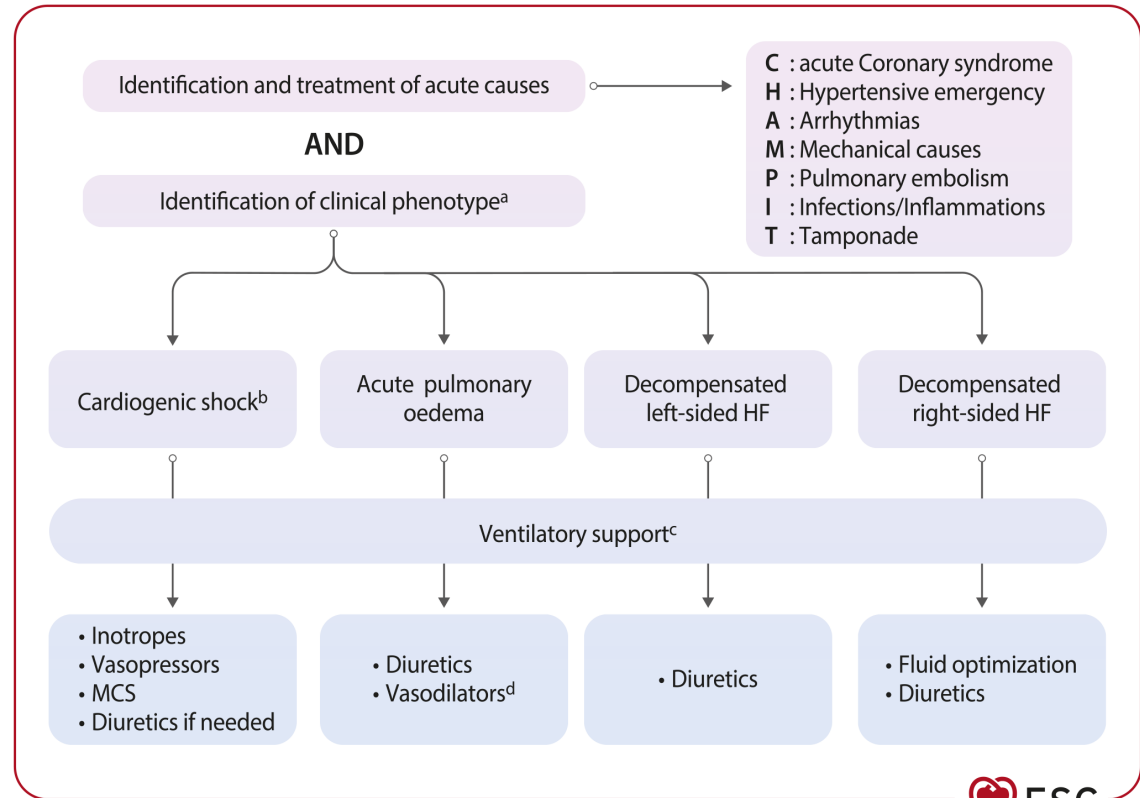
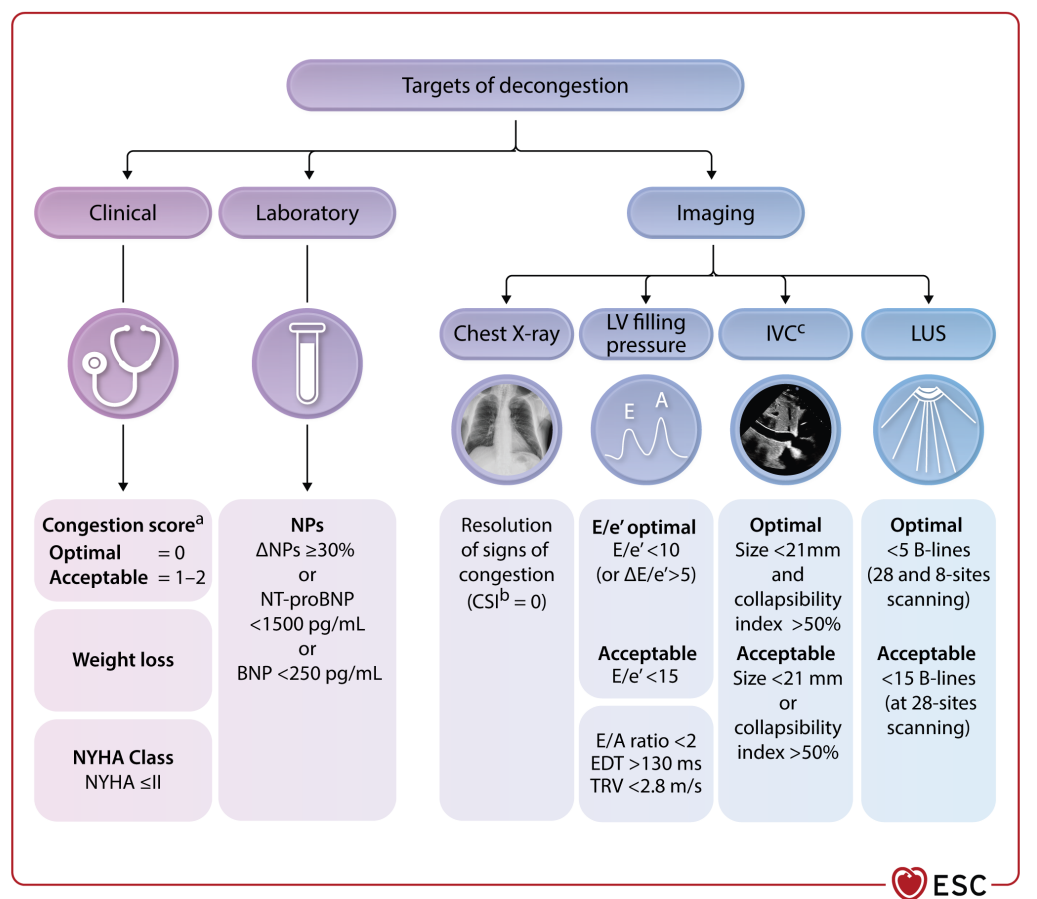


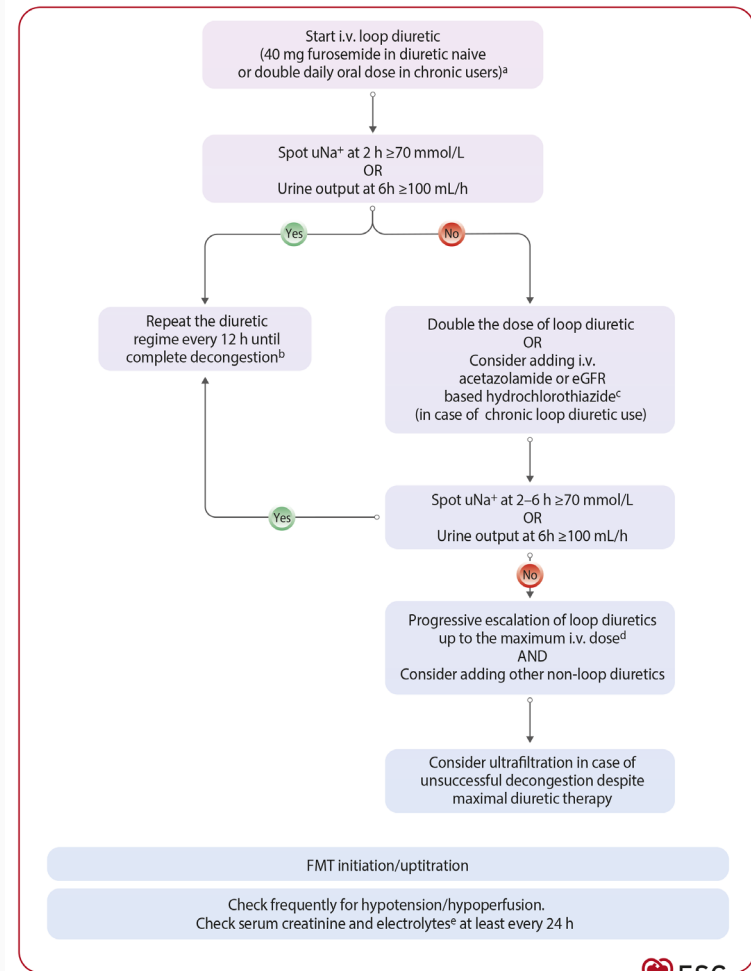
Figure 14 Tools used for assessment of decongestion during the pre-discharge phase



Recommendations for pre-discharge and early post-discharge follow-up of patients hospitalized for decompensated heart failure

Recommendations	Class	Level
Careful evaluation before discharge is recommended in patients hospitalized for DHF to exclude persistent signs of congestion.	I	C
An intensive strategy of rapid initiation and uptitration of FMT before discharge and during frequent follow-up visits in the first 6 weeks following an HFH is recommended to reduce the risk of HF rehospitalization or death.	I	B2

Figure 15 Management of decongestion



Recommendations for the immediate and intermediate treatment of decompensated heart failure (1)

Recommendations	Class	Level
Oxygen and ventilatory support		
Oxygen is recommended in patients with $\text{SpO}_2 < 90\%$ or $\text{PaO}_2 < 60$ mmHg to correct hypoxaemia.	I	C
Intubation is recommended in patients with DHF and persistent and progressive respiratory failure despite oxygen administration or non-invasive ventilation to correct hypoxaemia.	I	C
Non-invasive positive pressure ventilation should be considered in patients with respiratory distress (respiratory rate > 25 breaths/min, $\text{SpO}_2 < 90\%$) and started as soon as possible in order to decrease respiratory distress and reduce the risk of endotracheal intubation.	IIa	B2
Diuretics		
Intravenous loop diuretics are recommended for all patients with DHF admitted with fluid overload to improve symptoms.	I	A
The addition of short-term intravenous acetazolamide or oral hydrochlorothiazide should be considered in patients with fluid overload previously treated with loop diuretics, to reduce congestion.	IIa	B1
Urinary Na^+ -guided diuretic therapy may be considered during the first days of treatment of patients with DHF to improve natriuresis and diuresis.	IIb	B1
SGLT2-I		
In-hospital initiation of SGLT2-I is recommended in patients with DHF after initial stabilization to improve QoL and congestion symptoms and reduce the risk of HFH.	I	B1

Recommendations for the immediate and intermediate treatment of decompensated heart failure (2)

Recommendations	Class	Level
Vasodilators		
Intravenous vasodilators may be considered as initial therapy in patients with DHF and systolic blood pressure >110 mmHg to improve symptoms and reduce congestion.	IIb	C
Inotropic agents		
Inotropic agents may be considered in patients with systolic blood pressure <90 mmHg and evidence of hypoperfusion who do not respond to standard treatment, including fluid challenge, to improve perfusion.	IIb	C
Vasopressors		
Vasopressors, preferably norepinephrine, may be considered in patients with cardiogenic shock to increase blood pressure and vital organ perfusion.	IIb	C
Opiates		
Routine use of opiates is not recommended in patients with DHF unless severe/intractable pain or anxiety, due to risk of harm.	III	C

Recommendations for the use of temporary mechanical circulatory support in patients with cardiogenic shock

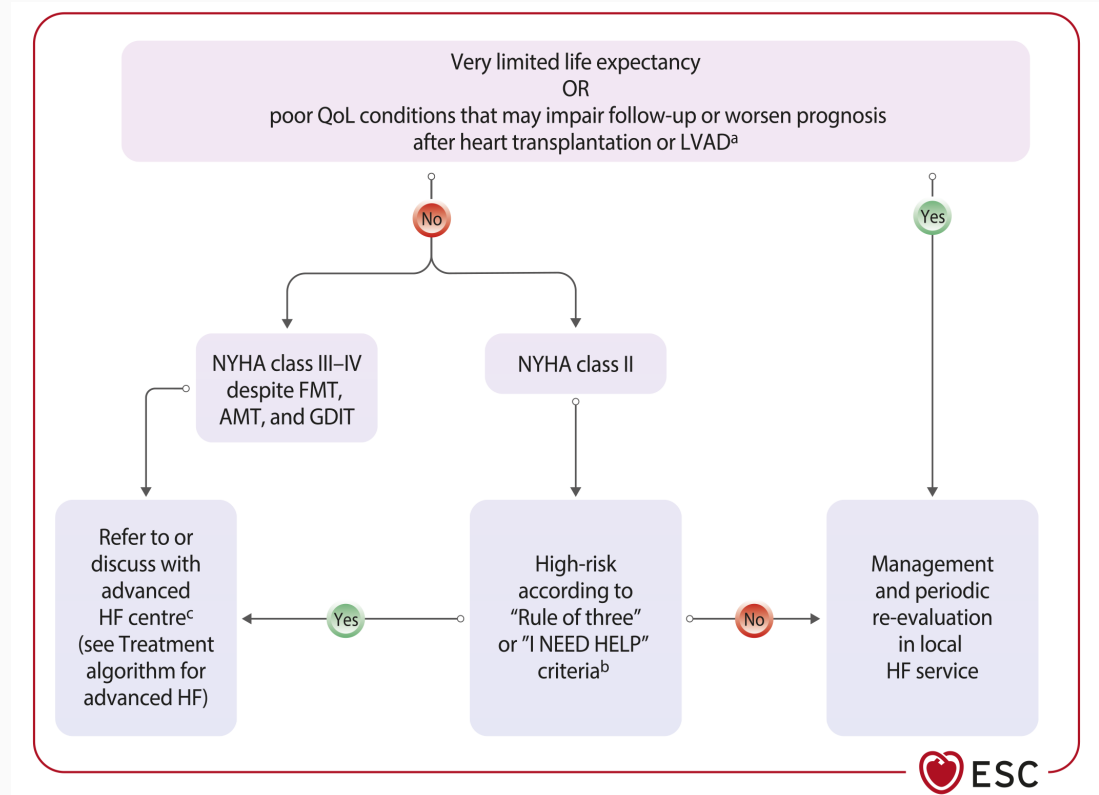
Recommendations	Class	Level
A multidisciplinary Shock Team is recommended in potential candidates for temporary MCS to guide the device selection (modality and type) based on patient and HF characteristics.	I	C
Cardiogenic shock caused by acute MI		
Temporary MCS with a microaxial flow pump should be considered in selected patients with cardiogenic shock caused by ST-elevation MI with left ventricular systolic dysfunction and no risk of hypoxic brain injury in order to reduce the risk of death.	IIa	B1
Temporary MCS is not recommended in unselected patients with cardiogenic shock caused by acute MI, due to risk of harm.	III	B1
IABP is not recommended in unselected patients with cardiogenic shock, due to the lack of effect.	III	B1
MI-related mechanical complications		
Temporary MCS should be considered in patients with mechanical complications related to MI as a bridge to definitive treatment.	IIa	C
Advanced HF cardiogenic shock		
Temporary MCS should be considered in selected patients with HF-related haemodynamic instability as a BTR, BTD, BTB, BTC, or BTT.	IIa	C

Criteria for definition of advanced heart failure

All the following criteria must be present despite FMT, AMT and GDIT:

1. Severe and persistent symptoms of HF (NYHA class III or IV).
2. Severe cardiac dysfunction defined by at least one of the following:
 - LVEF $\leq 30\%$
 - Isolated RV failure (e.g. ARVC)
 - Non-operable severe valve abnormalities
 - Non-operable severe congenital abnormalities
 - Persistently high (or increasing) BNP or NT-proBNP values and severe LV diastolic dysfunction or structural abnormalities (i.e. HFpEF, hypertrophic or restrictive cardiomyopathies).
3. Episodes of pulmonary or systemic congestion requiring high-dose i.v. diuretics (or diuretic combinations) or malignant arrhythmias causing ≥ 2 unplanned visits or hospitalizations in the last 12 months, or ≥ 1 episode of low output requiring inotropes or vasoactive drugs.
4. Severe impairment of exercise capacity with inability to exercise, or low 6-min walk test distance (< 300 m), or $pVO_2 < 12\text{--}14$ mL/kg/min, or $< 50\%$ predicted value, estimated to be of cardiac origin.

Figure 16 Triage and timely referral of patients with advanced heart failure



Criteria for high risk of advanced heart failure

Rule of three criteria (at least 1):

- HFH (≥ 2 or ≥ 1 requiring inotropes within the last year)
- Diuretic resistance
- Intolerance to neurohormonal drugs with onset of cardiorenal syndrome

I NEED HELP criteria (at least 1):

- Inotropes
- NYHA class III/IV or persistently high natriuretic peptides
- End-organ dysfunction (liver, kidney)
- EF $< 20\%$ or poor RV function
- Defibrillator shocks
- HFH > 1
- Escalating diuretics
- Low blood pressure, high heart rate
- Prognostic medication intolerance or the need to downtitrate

Recommendations for diagnosis and evaluation of advanced heart failure

Recommendations	Class	Level
Early consultation with an advanced HF centre is recommended in patients with advanced HF or at risk of advanced HF, who are motivated and do not have absolute contraindications for heart transplantation or durable MCS, in order to evaluate candidacy.	I	A
Cardiopulmonary exercise testing is recommended in patients with advanced HF, as part of the evaluation for heart transplantation and durable MCS.	I	C
Right heart catheterization is recommended in patients with advanced HF, as part of the evaluation for heart transplantation and durable MCS.	I	C

Figure 17 INTERMACS classification

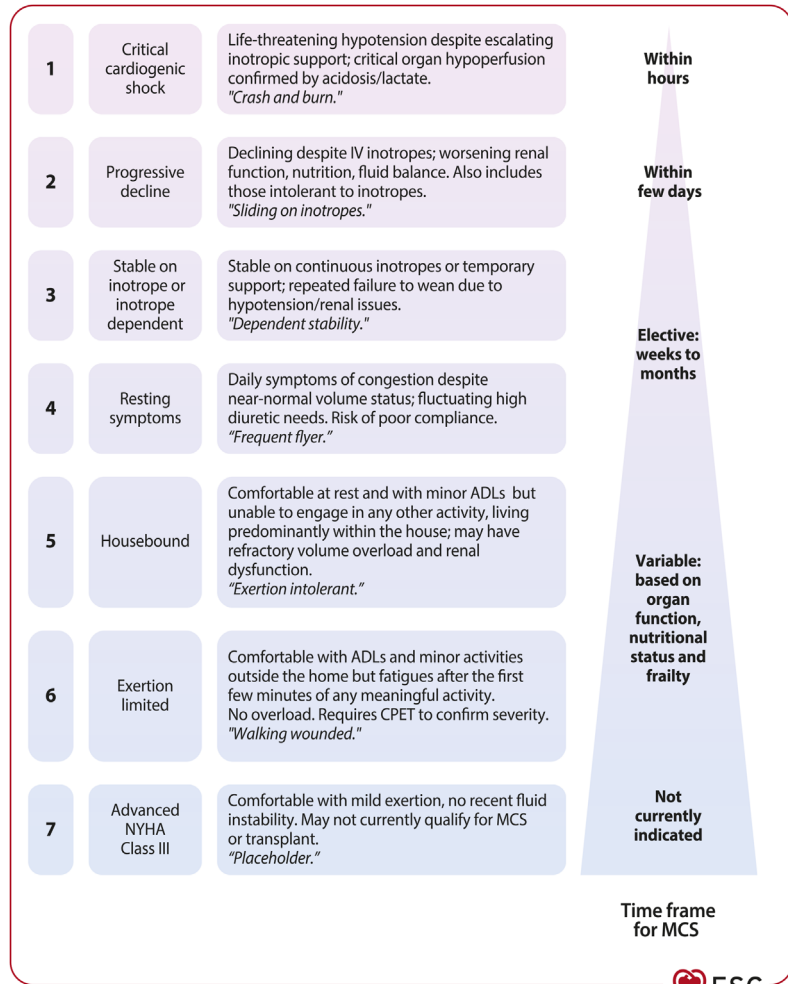
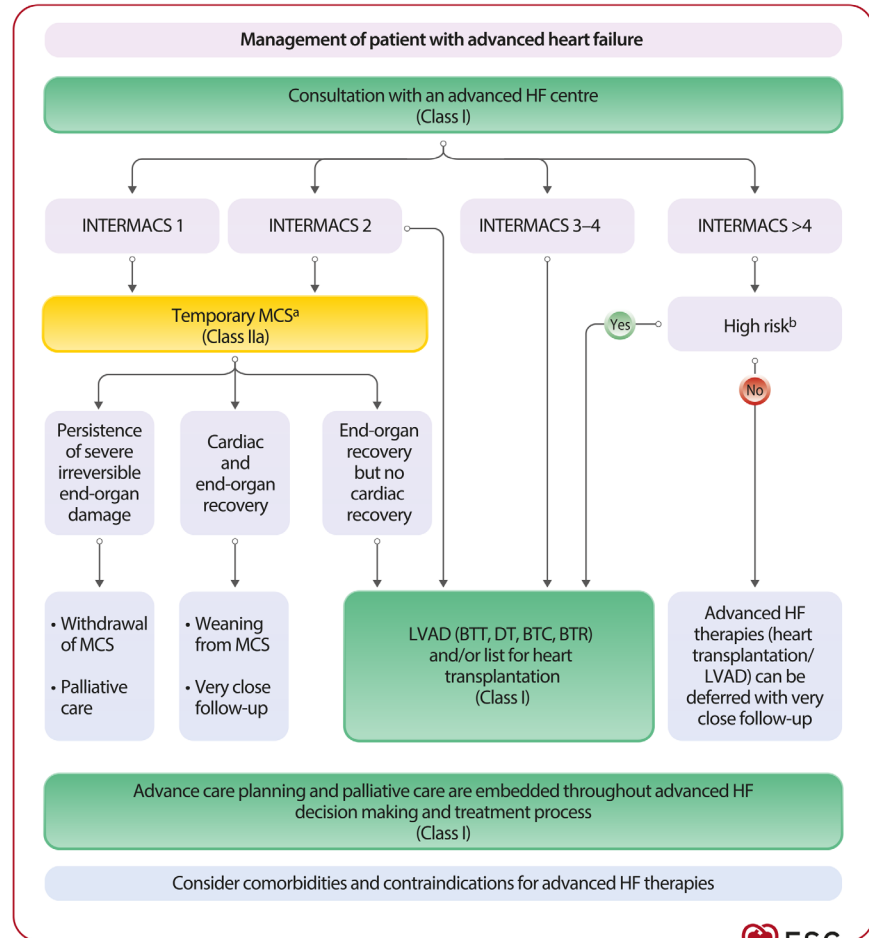


Figure 18 Treatment for advanced heart failure



Recommendations for medical management of advanced heart failure

Recommendations	Class	Level
Downtitration or discontinuation of beta-blockers or ivabradine should be considered in selected patients with advanced HFrEF and evidence of organ hypoperfusion, despite initial treatment, to increase cardiac output and improve symptoms.	IIa	C
Continuous inotropes should be considered in patients with advanced HFrEF, low cardiac output, and evidence of hypoperfusion or end-organ dysfunction, as BTD, BTT or as bridge to durable MCS, to increase cardiac output and improve symptoms.	IIa	C
Kidney replacement therapy should be considered in patients with refractory volume overload and end-stage kidney failure to improve symptoms.	IIa	C
Ultrafiltration may be considered in patients with refractory volume overload unresponsive to escalation of diuretic therapy to reduce the risk of rehospitalization.	IIb	C

Indications and contraindications for implantation of a left ventricular assist device (1)

Indications

HF with severely reduced LVEF and one of the following:

- INTERMACS 1-2 with neurological and other end-organ recovery but no cardiac recovery on temporary MCS
OR
- INTERMACS 2–4
OR

Severe symptoms and signs (NYHA class III/IV; INTERMACS >4) despite optimal FMT and GDIT, and at least one of the following:

- Unable to exercise due to HF or, if able to perform CPET, with $pVO_2 < 12\text{--}14 \text{ mL/kg/min}$ and/or $< 50\%$ predicted value.
- End-organ dysfunction (worsening kidney and/or hepatic function, group 2 pulmonary hypertension, cardiac cachexia) due to reduced perfusion as indicated by cardiac index $\leq 2 \text{ L/min/m}^2$ despite euvolaemic or therapy-resistant hypervolaemic status.
- Repeated HFH within the previous 12 months without an obvious precipitating cause.

Indications and contraindications for implantation of a left ventricular assist device (2)

Contraindications

Absolute:

- Severe right ventricular dysfunction despite euvolaemic status
- Severe irreversible end-organ (kidney and/or liver) disease
- Severe irreversible neurological disease
- Severe pulmonary disease
- Contraindication to long-term oral anticoagulation
- Medical non-adherence and/or other psychosocial limitations
- Insufficient social supports to achieve compliant care in the outpatient setting
- Psychological instability that jeopardizes proper follow-up and intensive therapeutic regime after implantation

Relative:

- | | |
|--|--|
| <ul style="list-style-type: none">• Obesity or malnutrition• Musculoskeletal disease that impairs rehabilitation• Malignancy (depending on type and prognosis)• Severe peripheral arterial disease• Systemic infection | <ul style="list-style-type: none">• Alcohol or substance abuse within the past 6 months, or active smoking• Impaired cognitive function• Frailty |
|--|--|

Recommendations for durable mechanical circulatory support in advanced heart failure

Recommendations	Class	Level
Durable MCS (LVAD) is recommended in selected patients with advanced HFrEF, despite FMT and GDIT, as BTT, BTC, BTR, or as destination therapy to improve symptoms and reduce the risk of death.	I	C
Histopathological examination of explanted heart tissue from LVAD or heart transplantation surgery should be considered in patients with non-ischaemic HF to identify an aetiological diagnosis that may be treatable or facilitate family screening.	IIa	C

Indications and contraindications for heart transplantation

Indications

Advanced HF and no other therapeutic option (except for durable MCS as BTT)

Absolute contraindications

Inability to comply with the therapeutic regimen (e.g. dementia)

Serious comorbidities with poor prognosis, not reversible with heart transplantation

Other contraindications

Active infection

Severe peripheral arterial or cerebrovascular disease

Elevated pulmonary vascular resistance (if pharmacologically irreversible)

Cancer

Irreversible liver dysfunction (cirrhosis) or irreversible kidney dysfunction (e.g. creatinine clearance $<30 \text{ mL/min/1.73 m}^2$)

Systemic disease with multiorgan involvement

BMI $>35 \text{ kg/m}^2$ or severe cachexia

Alcohol or substance abuse or active smoking within the past 6 months

Psychological instability

Insufficient social support to achieve compliant care in the outpatient setting

Recommendations for heart transplantation

Recommendation	Class	Level
Heart transplantation is recommended in selected patients with advanced HF refractory to FMT and GDIT, and without contraindications, to improve QoL and survival.	I	C

Updated definitions for the CHA₂DS₂-VA score

	Description	Point
C	Chronic HF stage B, C, D	1
H	Hypertension	1
A	Age ≥75 years	2
D	Diabetes mellitus	1
S	Stroke/transient ischaemic attack/arterial thromboembolism	2
V	Vascular disease (CAD and/or PAD)	1
A	Age 65–74 years	1

Recommendations for management of atrial fibrillation in patients with heart failure (1)

Recommendations	Class	Level
Recommendations for anticoagulation		
Oral anticoagulation is recommended in patients with clinical AF at elevated thromboembolic risk, as determined by CHA ₂ DS ₂ -VA score, to prevent ischaemic stroke and thromboembolism.	I	A
DOACs are recommended in preference to VKAs in patients with HF to prevent stroke and thromboembolism.	I	B1
Rate control		
Beta-blockers are recommended in stable patients with HFrEF and AF as first-line therapy for short- and long-term rate control.	I	C
Digoxin should be considered in stable patients with HFrEF and AF when the ventricular rate remains high despite beta-blockers, or when beta-blockers are contraindicated/not tolerated, in order to obtain short- and long-term rate control.	IIa	C
Atrioventricular node ablation combined with CRT should be considered in patients with severely symptomatic permanent AF and poor rate control despite medical therapy and at least one HFH to reduce symptoms, physical limitations, recurrent HFH and death.	IIa	B1
Intravenous amiodarone or digoxin may be considered in haemodynamically unstable patients with HFrEF and AF to stabilize the patient and achieve acute control of heart rate.	IIb	C

Recommendations for management of atrial fibrillation in patients with heart failure (2)

Recommendations	Class	Level
Cardioversion		
Urgent cardioversion is recommended in the setting of DHF in patients presenting with rapid ventricular rates and haemodynamic instability to restore sinus rhythm.	I	C
Atrial fibrillation catheter ablation		
Catheter ablation for AF should be considered in selected patients with symptomatic AF and HFrEF to improve QoL and reduce the risk of HFH or death.	IIa	C

Recommendations for revascularization in chronic coronary syndrome in patients with heart failure

Recommendations	Class	Level
Revascularization for symptom improvement (HFpEF and HFrEF)		
Coronary revascularization is recommended in patients with HF, obstructive CAD, and persistent angina, despite FMT for HF and antianginal drugs, to relieve symptoms.	I	C
Revascularization for outcome improvement (HFrEF)		
If revascularization is planned after assessment by a Heart Team, CABG should be considered as the preferred revascularization strategy, in patients with LVEF $\leq 35\%$ and multivessel CAD suitable for surgery, to improve long-term survival.	IIa	B1

Figure 19 Management of coronary artery disease in patients with heart failure

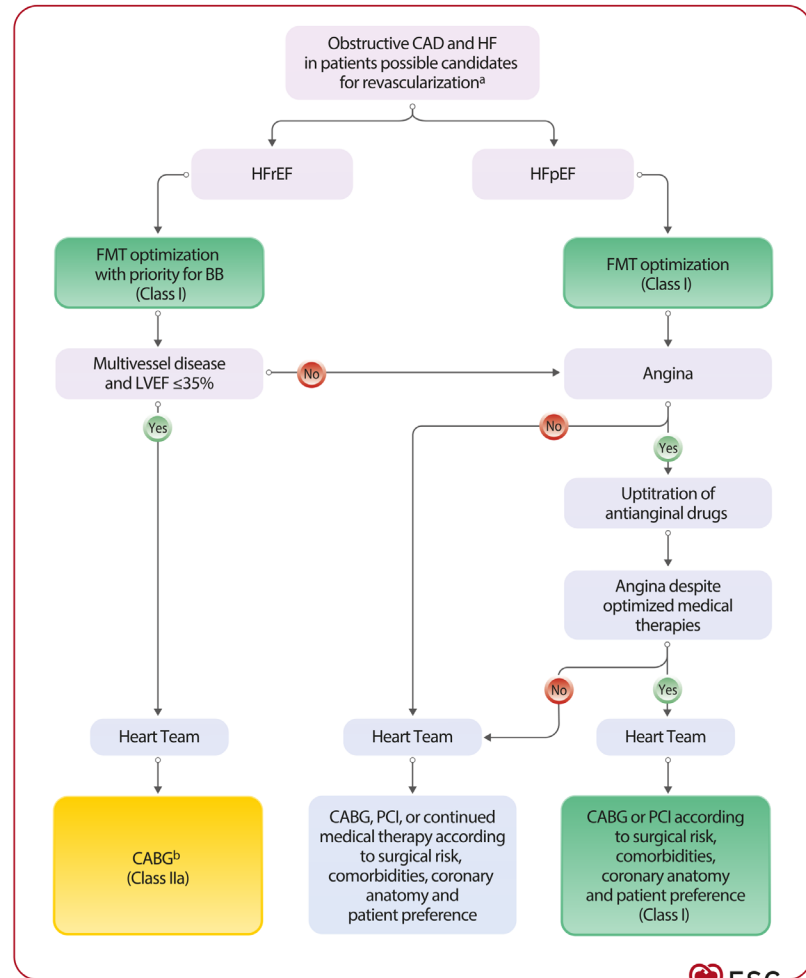
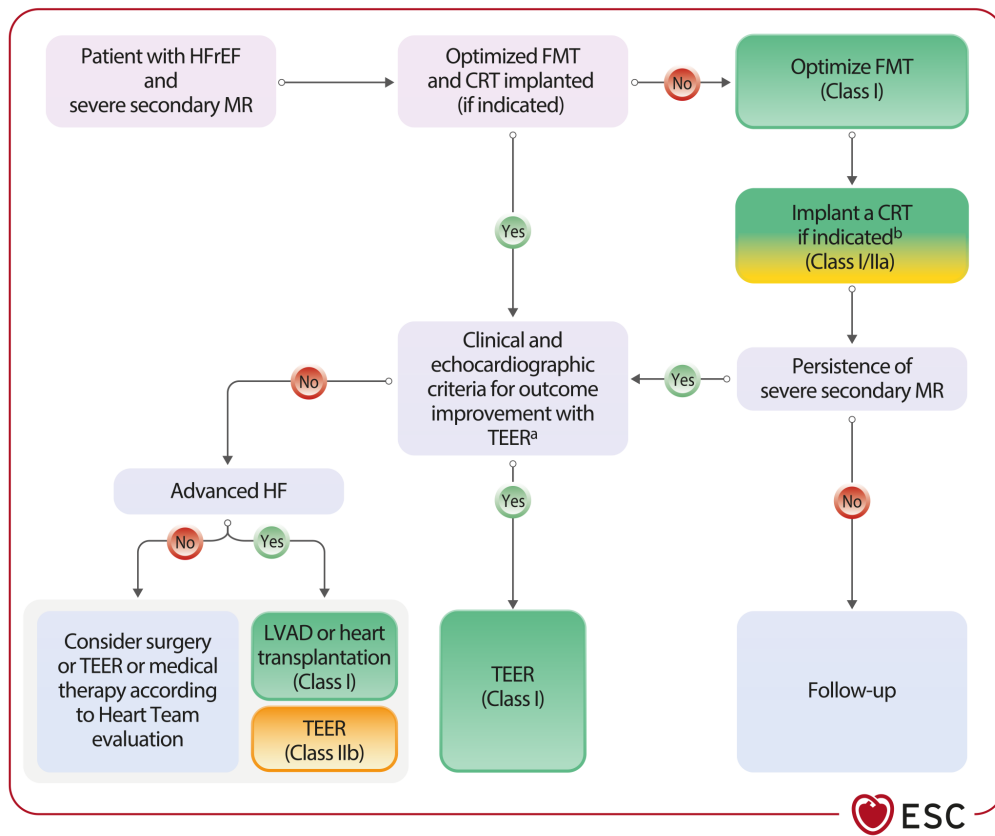


Figure 20 Management of secondary mitral regurgitation in patients with heart failure with reduced ejection fraction



Selection criteria for mitral transcatheter edge-to-edge repair in heart failure with reduced ejection fraction

Clinical and echocardiographic criteria

Optimized FMT and other GDIT (i.e. CRT) if needed

NYHA class $\geq$ II

At least one HFH within the last year

OR

Increased natriuretic peptide levels (BNP $\geq$ 300 pg/mL or NT-proBNP $\geq$ 1000 pg/mL)

No stage D (advanced) HF

No CAD requiring revascularization

No hypertrophic, restrictive, or infiltrative cardiomyopathies

LVEF 20–50%

LVESD $\leq$ 70 mm

Anatomy deemed suitable for mitral TEER

sPAP $\leq$ 70 mmHg

No severe RV dysfunction

No severe aortic and/or tricuspid valve disease

Recommendations for valvular heart disease in patients with heart failure

Recommendations	Class	Level
Aortic stenosis		
Aortic valve intervention, SAVR or TAVI, is recommended in patients with severe aortic stenosis and HF to improve symptoms and reduce the risk of death.	I	B1
Secondary mitral regurgitation		
Mitral transcatheter edge-to-edge repair is recommended in haemodynamically stable, symptomatic patients with HFrEF and persistent severe secondary mitral regurgitation despite optimized FMT and CRT if indicated, who fulfil specific clinical and echocardiographic criteria, in order to reduce the risk of HFH and improve QoL.	I	B1
Mitral transcatheter edge-to-edge repair may be considered in selected symptomatic patients with HFrEF and persistent severe secondary mitral regurgitation despite optimized FMT and CRT if indicated (after excluding the option of LVAD or heart transplantation) who do not fulfil the clinical and echocardiographic criteria for outcome improvement, in order to improve symptoms.	IIb	C

Recommendations for obesity in patients with heart failure

Recommendations	Class	Level
Semaglutide or tirzepatide should be considered for patients with symptomatic HF, LVEF $\geq 45\%$, and BMI ≥ 30 kg/m ² , regardless of diabetes status, to reduce body weight, and improve exercise capacity and QoL.	IIa	B1
Bariatric surgery may be considered in obese patients with HF and a BMI ≥ 35 kg/m ² to reduce body weight when repetitive and structured lifestyle changes combined with weight-reducing medications do not result in maintained weight loss.	IIb	C

Recommendations for iron deficiency in patients with heart failure

Recommendations	Class	Level
Intravenous iron supplementation is recommended in patients with symptomatic HFrEF and iron deficiency to alleviate HF symptoms and improve QoL.	I	B1
Intravenous iron supplementation should be considered in patients with symptomatic HFrEF and iron deficiency to reduce the risk of HFH.	IIa	B1

Recommendations for lung disease and sleep-disordered breathing

Recommendations	Class	Level
Adaptive servo-ventilation may be considered in patients with HFrEF and sleep-disordered breathing with predominant obstructive sleep apnoea to improve sleep quality, health-related QoL and symptoms.	IIb	C
Adaptive servo-ventilation is not recommended in patients with HFrEF and sleep-disordered breathing with predominant central sleep apnoea because of an increased risk of CV and all-cause death.	III	A

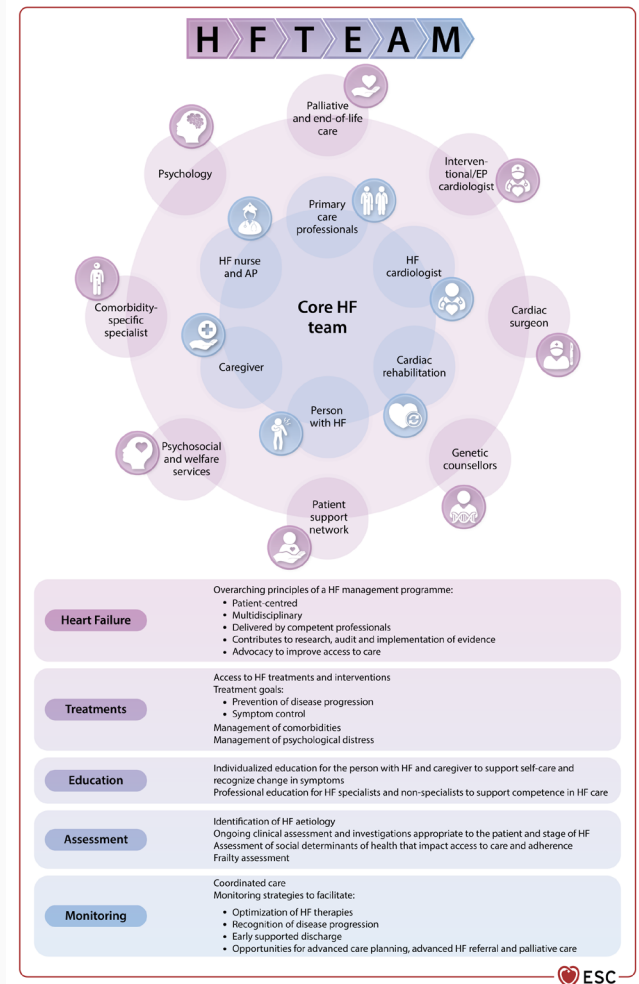
Recommendations for anxiety, depression, and frailty

Recommendation	Class	Level
Assessment of anxiety, depression and frailty should be considered in patients with HF to support the development of personalized care plans and to identify factors that may contribute to adverse outcomes.	Ia	C

Recommendations for other non-cardiovascular comorbidities

Recommendations	Class	Level
Influenza and pneumococcal vaccination should be considered in patients with HF to reduce the risk of pneumonia complications, hospitalization, and death.	IIa	C
NSAIDs and COX-2 inhibitors are not recommended in patients with HF as they increase the risk of HF worsening and HFH.	III	B2

Figure 21 Core members of the heart failure team with expanded team members depending on patient needs and local availability



Recommendations for the multidisciplinary management of heart failure

Recommendations	Class	Level
A multidisciplinary HF management programme is recommended in patients with HF to reduce the risk of HFH and death.	I	B1
HF education and self-management strategies in patients with HF are recommended to reduce the risk of HFH or death.	I	A
Multidisciplinary interventions to improve adherence, including assessment of health literacy and polypharmacy review, are recommended in patients with HF to reduce the risk of hospitalization and death.	I	B2

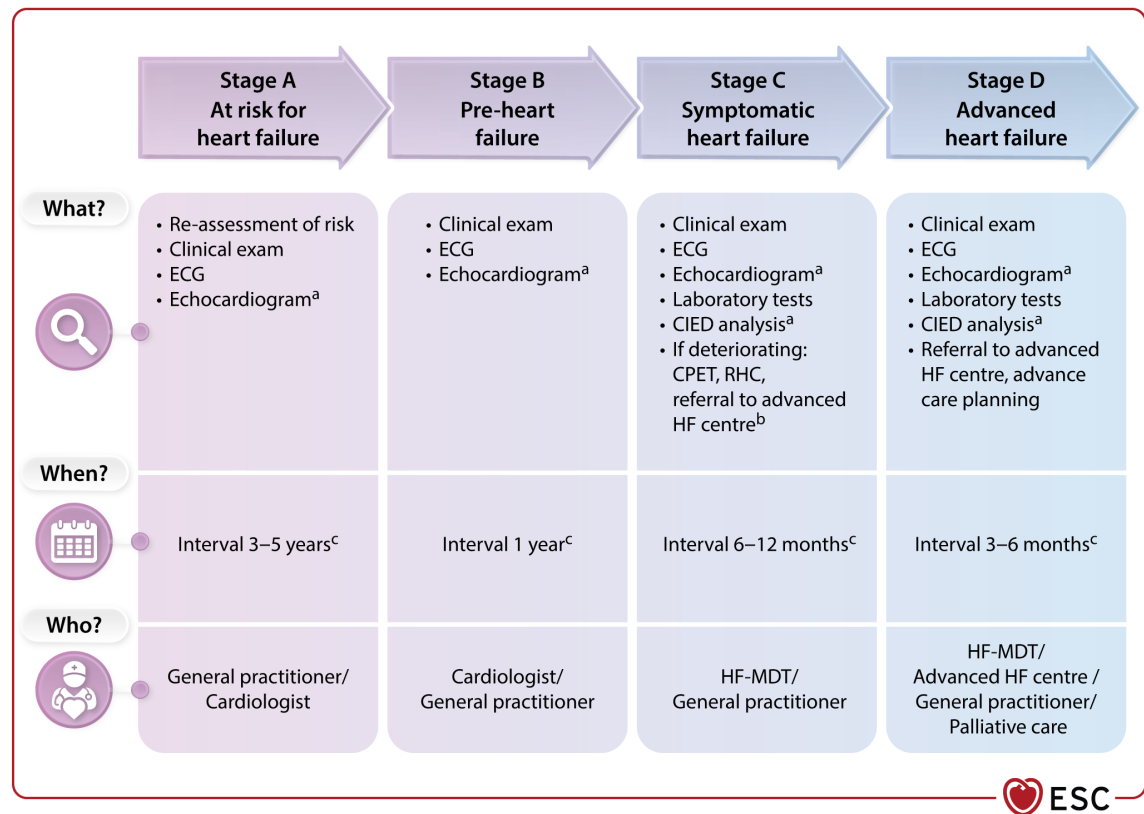
Recommendations for exercise training and cardiac rehabilitation

Recommendations	Class	Level
Personalized exercise training, in the context of multidisciplinary exercise-based cardiac rehabilitation, is recommended for all stable patients, unless there are specific contraindications, in order to improve exercise capacity and QoL, and reduce the risk of all-cause hospitalization.	I	B1
Personalized exercise training outside exercise-based cardiac rehabilitation is recommended for all stable patients on a long-term basis, unless there are specific contraindications, in order to improve exercise capacity and QoL, and reduce the risk of all-cause hospitalization.	I	B2
Cardiac rehabilitation is recommended in patients with an LVAD to improve functional capacity and QoL.	I	B1
Cardiac rehabilitation should be considered in patients with DHF, as soon as safely possible, to improve functional capacity and QoL.	IIa	B1

Recommendations for invasive and non-invasive telemonitoring

Recommendations	Class	Level
Remote haemodynamic monitoring of PA pressure should be considered in symptomatic patients with HF, NYHA class III and a HFH in the past 12 months in order to reduce the risk of HFH.	IIa	B1
Non-invasive telemonitoring modalities may be considered for patients with HF in order to reduce the risk of HFH.	IIb	B1

Figure 22 Long-term follow-up of patients with stable heart failure – what – when – who



Recommendations for long-term follow-up

Recommendations	Class	Level
Proactive discussion regarding HF trajectory, goals of care, and advance care planning is recommended in patients with advanced HF to facilitate communication on end of life and QoL.	I	B1
Access to an integrated HF palliative care multidisciplinary team is recommended for patients in an advanced HF stage to improve QoL and reduce symptom burden.	I	B2

Recommendations for heart failure and pregnancy

Recommendations	Class	Level
Pre-conception care and counselling is recommended for patients with HF to facilitate decision-making surrounding HF treatments, pregnancy, contraception, pre-implantation genetic screening, and assisted reproductive therapies.	I	C
In pregnant women with HFrEF, it is recommended that non-selective beta-blockers are switched to beta1-selective blockers (metoprolol, bisoprolol) with close mother and foetus monitoring to reduce the risk of adverse foetal events and a lower birth weight.	I	C
ACE-Is, ARBs, ARNIs, MRAs, ivabradine, and SGLT2-Is are not recommended during pregnancy due to the risk of foetotoxicity or teratogenicity.	III	C

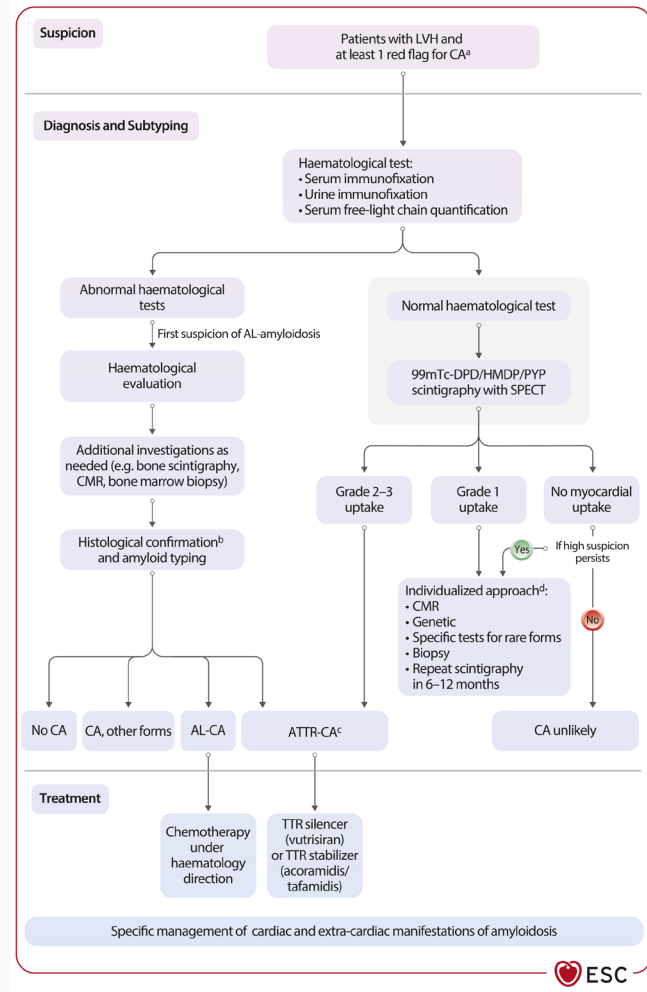
'Red flags' for most common forms of cardiac amyloidosis (1)

Category	Red flag	TTR	AL
Extracardiac	Polyneuropathy	X	X
	Dysautonomia	X	X
	Skin bruising	–	X
	Macroglossia	–	X
	Deafness	X	–
	Bilateral carpal tunnel syndrome	X	–
	Ruptured biceps tendon	X	–
	Lumbar spinal stenosis	X	–
	Vitreous deposits	X	–
	Family history	X	–
	Kidney insufficiency	X	X
	Proteinuria	X	X

‘Red flags’ for most common forms of cardiac amyloidosis (2)

Category	Red flag	TTR	AL
Cardiac clinical	Hypotension or normotension if previously hypertensive	X	X
ECG	Pseudo-infarct ECG pattern	X	X
	Low/decreased QRS voltage relative to LV thickness	X	X
	AV conduction disease	X	X
Laboratory	Disproportionally elevated NT-proBNP relative to HF severity	X	X
	Persistently elevated cardiac troponin levels	X	X
Echocardiography	Granular sparkling of myocardium	X	X
	Increased right ventricular wall thickness	X	X
	Increased valve thickness	X	X
	Increased interatrial septum thickness	X	X
	Pericardial effusion	X	X
	Reduced longitudinal strain with apical sparing pattern	X	X
CMR	Global subendocardial or transmural LGE	X	X
	Elevated native T1 values	X	X
	Increased extracellular volume	X	X
	Abnormal gadolinium kinetics	X	X

Figure 23 Diagnosis and treatment of cardiac amyloidosis in heart failure patients



Recommendations for cardiac amyloidosis

Recommendation	Class	Level
Treatment with a TTR silencer (vutrisiran) or stabilizer (tafamidis or acoramidis) for patients with variant or wild-type transthyretin cardiac amyloidosis and NYHA class I–III is recommended to reduce progression of symptoms and the risk of CV hospitalization and all-cause death.	I	A